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SMEINST-07-2016-2017

TITLE

CLOUD-ENABLED IR SPECTROSCOPY INTEGRATION PLATFORM FOR CUSTOMISATION OF AFFORDABLE, REAL-TIME CHEMICAL MONITORING SOLUTIONS IN THE FOOD AND PHARMACEUTICAL INDUSTRIES

ACRONYM

visum LAB TO LINE

PARTICIPANT LIST

Participant no.	Participant organisation name	Country
1 (Coordinator)	VISUM Ltd.	Ireland

TABLE OF HISTORY OF CHANGES

DoA version and date	Description of changes	Page
V2-08.06.2016	The following section has been added: 4.3 SUBCONTRACTING- BEST VALUE FOR MONEY	31
V3-31.05.2018	As per the Amendment letter (dated 31.05.2018 In view of a requested 3-month extension to the project duration and the changes in numbers of potential clients participating in the project and the level at which they are involved, edits have been made in Pages 4, 5, 7, 10, 11, 13, 15, 19, 21, 22, 23, 25, 26, 27, 28, 29, 30 of Part B as follows: 24-month workplan has been changed to 27-month workplan Delivery date of D3.3 has been changed from M22 to M24 Delivery date of D3.4 has been changed from M24 to M27 Delivery date of D4.3 has been changed from M23 to M26 Delivery date of D4.4 has been changed from M24 to M27 Delivery date of D5.3 has been changed from M22 to M25 Delivery date of D5.4 has been changed from M24 to M25 Delivery date of Milestone 9 has been changed from M22 to M26 Delivery date of Milestone 10 has been changed from M23 to M26 Gantt chart has been revised to reflect the above changes In addition, references to the original four early adaptors have been replaced to reflect that there are now seven early adaptors, five of whom are participating in the project as end users. The associated changes in implementation and products of interest are incorporated in the amended document. In the Section 4.1, the CVs of the key personnel have been updated. In addition to the above changes, IRIS Advanced Engineering Limited has been rebranded to VISUM Ltd. This change in name is implemented throughout All the above changes have also been implemented as relevant in the Part A Workplan Tables.	4 5 7 10 11 13 15 19 21 22 23 25 26 27 28 29 30

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1. EXCELLENCE

1.1 OBJECTIVES

INTRODUCTION TO THE BUSINESS NEED

Over the past 5 to 10 years we have seen important advances in Infrared (IR) spectroscopy technology, with the arrival on the market of low-cost, miniature, fit-for-purpose sensors that can replace bulky systems in R&D, in the field, and in manufacturing environments. While such sensor advancements are undoubtedly helping to increase industry use of NIR-based systems for in process monitoring, **the key to unlocking the floodgates for mainstream adoption lies in facilitating the successful integration of IR sensors into processing lines** and seamlessly providing the acquired process data to industry in the form of knowledge or intelligence.

Let's consider first of all the hardware integration. This is not just general engineering; this is specialised optical engineering and **proper sampling**. Sampling is key to the ability to take accurate readings from the material being processed and basically means optimising the interaction between light and sample in order to draw relevant information free of interferences and artefacts. If the sampling fails, you have fallen at the first hurdle irrespective of the performance of the spectrophotometer. There are no general solutions for sampling; cost-effective trade-offs cannot reliably tackle all the practical cases unless the sampling devices are being prepared by-design to be adapted to a board variety of practical cases.

And then there's the whole data side of things. Robust mathematical tools (i.e. chemometric modelling) are absolutely vital for **converting readings into accurate measurements**. While a major advantage of NIR spectroscopy technology is its **sensitivity to the sample matrix**, this same advantage makes NIR difficult to work with in that measurements are affected by changes in the sample matrix related to particle size, density, humidity, and temperature. Moreover, if we consider the context of food matrices and pharmaceutical excipients, these can be affected by seasonal, geographical, and other sources of change that can result in differences in the physicochemical compositions of the material, which can have a direct impact in the NIR spectral features. Critically, the development of algorithms based on an understanding of the absorption and scattering behaviours of diffuse reflectance and transmittance can enable the removal of matrix effects and improved model robustness.

The market reality is that today there are not so many players who are specialising in the development and customisation of fully integrated and customised in-process monitoring solutions and that can deal with the installation challenges that the technology is facing, such calibration of the optical device, addressing fouling, the influence of external factors affecting the system such as temperature, etc.; we tend to have instrument providers on the one side and manufacturers and processors on the other. **So there is a clear 'integration gap'**. Industries willing to take the step to shift their laboratory analysis to the line (and this step involves an almost philosophical and cultural shift in the company, as opposed to merely a technological investment), cannot reasonably do so without a reliable integration, customisation and maintenance service that. A typical food or pharmaceutical manufacturer (irrespective of their size) will not be equipped with the in-house engineering and programming team to get NIR spectroscopy sensors reliably integrated and deployed in their processing lines. In fact, there is a general lack of availability of engineers and scientists with sufficient expertise working with spectroscopy. Consequently, companies that want to use NIR cannot find qualified people to help them do it.

Our vision through the VISUM® platform is to bridge this integration gap. Our game changing flexible hardware and software (figure 1 below, and the architecture is described in Table 4 in Section 1.3) equips engineers that do not require a specialization in spectroscopy in the food, pharmaceutical and process industries to perform the rapid, easy, and cost-effective integration of robust, bespoke solutions that are reliable and stable in manufacturing and processing environments. The VISUM® platform is based on an 'integrated-by-design' approach that has been born out of a mission to unblock the bottlenecks of the cost and complexity of integration of IR-based 'mini-PAT' solutions in processing lines.

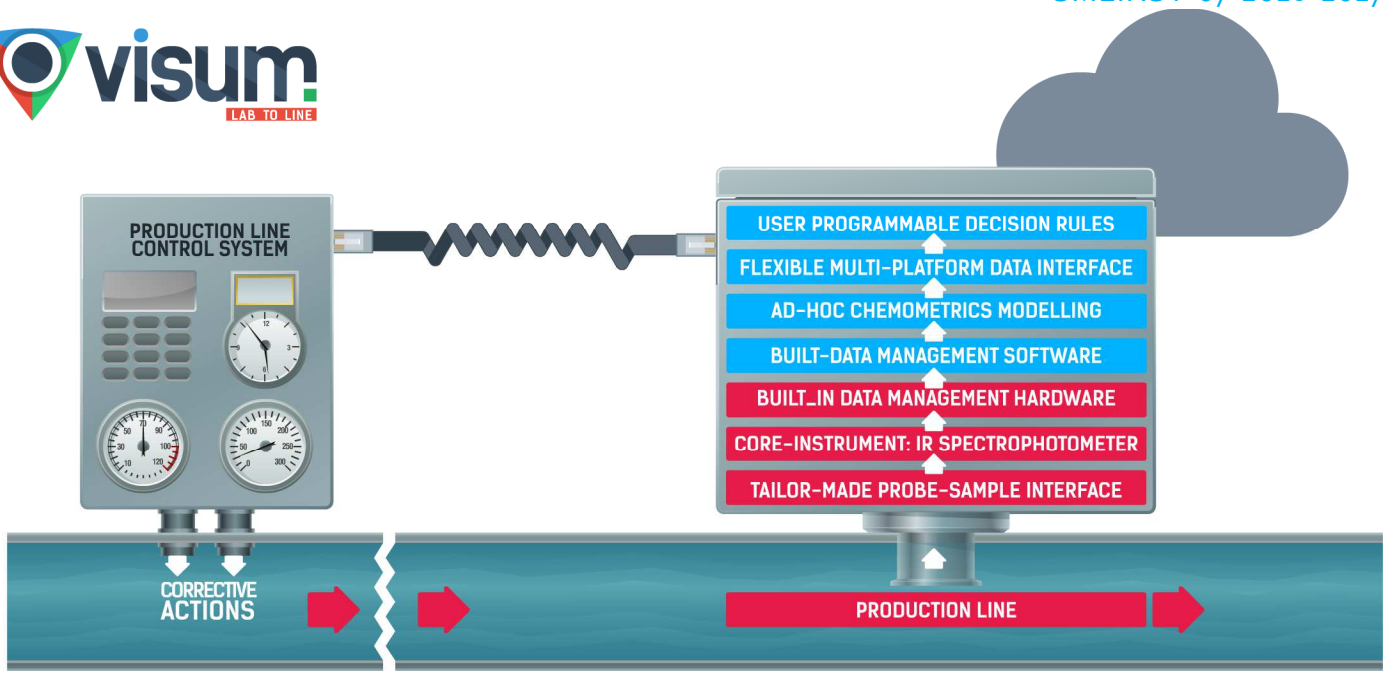

FIGURE 1- THE VISUM PLATFORM

We are on the edge of a wave that is about to take the industry by storm; that will see many applications that were formerly reserved for research-grade instruments being transferred to real-time, in-line monitoring and intelligent process control. The VISUM® platform will enable industries (large and small) from the food, pharmaceutical and other process sectors, to successfully ride this wave and benefit from greater process efficiencies (and cost savings), higher quality products, reduce waste and environmental impact.

SPECIFIC OBJECTIVES OF THE PROJECT

Building on the successful outcome of our Phase 1 project, we are now moving on to the Phase 2 SME Instrument, with the aim of accelerating the final stages of development and gearing up for the commercialisation of an **affordable IR-spectroscopy-based platform for enabling the rapid, easy integration and customisation of in-process chemical monitoring solutions in the food and pharmaceutical industries, resulting in improved process control, product quality, and safety**.

As such, the specific objectives that will be fulfilled over the course of a 27-month work plan are as follows:

1. **To build 5 application cases for the VISUM product in the dairy, snack food, meat and pharmaceutical** industries.
   The cooperation of test customers allows us to have an application-oriented product development as follows:
   a) ALMIRALL: Pharmaceutical sector (secondary production). An inline installation on a blender to monitor the mixing process for of the APIs and predict when the process end point is reached. The expected (required) precision is as low as 2% of nominal value.
   b) DAIRYGOLD: Dairy products. An inline installation in a pipe will continuously monitor specific composition of dairy products.
   c) METALQUIMIA: Snack products. Installation in a dry oven monitoring the moisture content of produce on a feed through conveyor belt.
   d) SALGOT: Meat based products. An inline system installed at the output of the stuffer line. Target parameters: Moisture, fat content and protein content
   e) DEC Pharma/ food processing equipment provider. End point of blending process defined by this end user's clients.

   Performance indicator: By M3, the 4 application cases in dairy, olive oil, meat and pharmaceuticals are defined, along with the respective integration strategies for the VISUM in each of the 4 industry facilities. Key Performance Indicators (KPIs) for validating the success of the industry trials are agreed with each test site.
2. **To develop the chemometric predictive models for each application.**
   Using the feasibility prototype system in VISUM, initial tests with product samples sets from our four beta testing clients will be performed for obtaining test data for the development project and for model development. An initial set of lab-scale ready-to-use predictive chemometric models will be developed, validated, and pre-installed in the respective VISUM units to be validated. Promising not widely already adopted chemometric methods (MCR -multivariate curve resolution- and MLR –multivariate linear regression- improved with wavelengths selection pre-treatment –genetic algorithms- as well as robust classification methods like SVM –support vector machines-) will be implemented for improving accuracy in relation to standard methods like Partial Least Squares (PLS) or PLS-DA, for example. ***Performance indicator:*** By M7, digital files for the first versions of each model are ready for laboratory validation and by M9, optimised versions are delivered for coding into the system software.

3. **To design and build the platform hardware layers**
   In keeping with the VISUM architecture (figure 1 and Table 4) and the system specifications that were defined in the Phase 1 SME Instrument project, the following hardware (HW) layers will be designed and built: a) Tailor-made probe-sample interfaces improved with "anti-fouling" windows; b) Standard customisable plug-in sampling unit for on-line version of VISUM for cases whereby the antifouling window is not suitable; c) Core instrument: IR spectrophotometer (whereby VISUM will be designed for versatility to enable compatibility with a variety of commercially available SWIR OEMs); d) System casing suited for 'click-in' integration of different IR spectrophotometers (depending on the measurement requirements of the application) as well as built-in computer that is capable of communicating with a selected variety of spectrophotometers firmware; e) Built-in data management HW. ***Performance indicator:*** By M4, an up to date portfolio of OEM SWIR spectrophotometers is complete so that we can ensure the VISUM designs are compatible with each; By M6 a dossier with the designs of the Standard probe-sample interfaces, the on-line sampler, housing and built in data management hardware is ready. By M10, HW units are built and ready for integration with system software.
4. **To design and develop the platform software layers and to build the cloud infrastructure**
   Again, in keeping with the VISUM architecture and the system specifications that have been defined in the Phase 1 project, the following software (SW) layers will be designed and developed: a) Built-in data management software; b) Ad-hoc chemometric models development environment; c) Flexible multiplatform data interface; d) User programmable decision rules, offering mini-PAT (Process Analytical Technology) capability. In keeping with the Industry 4.0 paradigm, the VISUM will be cloud enabled to allow the option of connecting sensor data to the cloud (as required) so that historical data can be analysed (Big Data), along with a web application (that will be accessible over all mobile platforms) to allow remote monitoring of networks of installed VISUMs. ***Performance indicator:*** By M10, the beta SW including the modelling code has been validated in the lab and debugging confirms readiness for integration with the system HW. By M11, the cloud database and Application Programming Interfaces are built.
5. **To integrate the system hardware and software and to perform in-house testing to ensure proper functioning of the integrated VISUM and effective communication with the cloud.** ***Performance indicator:*** By M12, tests with the integrated system confirm readiness for industry installation work to commence- Factory Acceptance Test (FAT) according to ISO 2001:9008 is approved.
6. **To install and test the VISUM system in the processing facilities of our participating test customers**.
   The VISUM system will be installed at the following end user sites: a) SALGOT- meat processing (to monitor salamis and sausages filling mixtures at the output of the stuffer line); b) DAIRYGOLD-dairy processing (to monitor milk quality parameters as flows through the pipes); c) METALQUIMIA-snack food production (to determine moisture content in a drying process) and, d) ALMIRALL- pharmaceutical manufacturing (in a blender), e) DEC- pharmaceutical manufacturing (in a blending line in order to validate/optimise the pre-defined integration strategies and confirm the ease of rapid integration (which is a key selling proposition of the VISUM) Production trials will be performed in order to validate the transferability of the calibration models and the performance and reliability of the VISUM technology vs. conventional monitoring methods. The VISUM will be interfaced with the existing process control system for the implementation of closed loop control either by means of the VISUM built-in control routines or using the existing control routines in the machinery (PLCs). ***Performance indicator:*** By M22, the systems are successfully installed at SALGOT, DAIRYGOLD, METALQUIMIA, ALMIRALL and DEC. By M22, correct operational performance is confirmed according to the pre-established KPIs and Site Acceptance Test (SAT) approval is obtained from each of the sites. By M27, the impacts of monitoring and controlling the process with the VISUM in-line system are quantified.
7. **To translate the validated prototype designs to product** through the creation of the manufacturing documentation package using the engineering information and specifications and to establish the manufacturing. ***Performance indicator:*** By M19, the product manufacturing documentation packages are passed to purchasing and production engineering to enable manufacture. By M23, the first 5 product units pass the quality assurance tests.
8. **To further develop the VISUM Business Plan and Business Models** for the commercialisation of the instrumentation (hardware) and data services. ***Performance indicator:*** By M22, our Business Plan is investor ready and we have fully detailed and verified our business models for realising the full potential of our VISUM platform in the European and global food, pharmaceutical and other markets.
9. **To gear up to market-** This will involve establishing and training the VISUM Sales Team and Application Team (for supporting the Sales Team), the finalization of the pricing structure and policy, and establishing the commercial/distribution networks for delivering on the sales targets in the key European and international markets. ***Performance Indicators:*** By M22 the Application Team is in place and trained, the first tier of our distribution network is in place and our pricing has been validated for timely progression to the Commercialisation Phase.

PROBLEM OR BUSINESS OPPORTUNITY AND HOW THE SOLUTION SOLVES THE PROBLEM OR AVAILS OF THE BUSINESS OPPORTUNITY

In line with the Industry 4.0 paradigm, there is increasing industry interest in shifting from laboratory-based or post-production quality

control to in-line monitoring of product and process parameters during processing. The main drivers for this interest are the **need to improve product quality (or to obtain standardised quality) and safety, to increase process efficiencies, to reduce waste and off spec product discard**, as well as increasing awareness of the **imperative of ICT adoption to remain competitive in our new Digital data-driven era**. There is an enormous business opportunity for providing even small and mid-tier enterprises with access to rapid and reliable in-process detection techniques that can keep pace with modern processing speeds and throughputs and which are suitable for: (i) assembly in the production line and operating under a realistic environment; (ii) enabling early detection of possible failures; (iii) performing permanent monitoring of the conditions; (iv) providing an assessment of conditions at any desired time[1]. The introduction of spectroscopic sensors for real-time process monitoring is enabling industry to gradually move from inferential monitoring and control toward continuous measurement of core quality parameters[2].

VISUM, in line with the objectives of H2020's Leadership in Enabling Technologies (LEIT) work programme, leverages photonics and ICT KETs (Key Enabling Technologies) to drive forward the **PAT** (Process Analytical Technologies) paradigm within the wider Industry 4.0 framework for improved industrial competitiveness.

Over the past 30 years, **on/in-line near infrared spectroscopy (NIRS)** (based on the absorption of electromagnetic radiation at wavelengths in the range 780–2500 nm) has proven to be one of the most efficient and advanced tools for continuous monitoring and control of process and product quality in the food processing industry.[3] NIR spectra of foods and pharmaceuticals comprise broad bands arising from overlapping absorptions corresponding mainly to overtones and combinations of vibrational modes involving C–H, O–H, and N–H chemical bonds[4], making it very feasible for measurements to be made in organic and biological systems such as meat, fish, fruit and vegetables, grain and grain products, dairy products, oils, beverages, Active Pharmaceutical Ingredients (APIs) and excipients, among others. However, despite the operating cost of NIRS being low, to date **the instruments themselves have been costly**, which has limited practical application[3]. Moreover, **calibration models based on NIR spectroscopy, and especially for on-line applications, tend to require a significant economical effort and are not reliable and stable enough when used practically**[3].

Our VISUM platform breaks down the cost and complexity barriers of NIR technology adoption for real-time process monitoring applications, by providing food and pharmaceutical processors (irrespective of their size or technological capability) with access to the necessary flexible hardware and software to allow for rapid, easy, and cost-effective integration of robust, customised IR solutions that are reliable and stable in manufacturing and processing environments. The aim is to ensure that integration can be performed even by non-specialised engineers. This is a timely development in view of the emergence of new compact, highly affordable IR spectroscopy sensors, such as those that are offered by Finnish technology company, Spectral Engines OY, which can be purchased for <€1k. Their business models are centred selling high volumes of sensors. The key now to mass adoption lies in facilitating the integration of these sensors in industry. (It is important to highlight that since we have commenced the VISUM development we have witnessed dramatic price drops for IR sensors from €5/6 to just under €1k- for orders of 10 units)

The VISUM platform will capitalise on this business opportunity by bringing to market the first **IR spectroscopy integration platform** that offers the following Unique Selling Propositions (USPs):

- **Affordability-** the use of the new wave of extremely affordable MEMS-NIR sensors to enable disruptive cost reductions (~up to 4-5 times cheaper) for fully customised in-process monitoring systems.
- **Ease of integration and reliability-** In relation to the reliability of the calibration models, the VISUM platform will incorporate a robust multivariate-models development environment and a clever 'calibration transfer program', thus eliminating the need for a full recalibration. This will be a fundamental feature in making the VISUM platform readily customisable for reliable use in many analytical measurement scenarios in industry.
- **Low maintenance-** A major practical problem that continues to pose great challenges in relation to in-line spectroscopy is biofouling and contamination of the optical window, caused by the environment in which the device is operating. VISUM offers two solutions (depending on the requirements of the application) for ensuring resistance to fouling- and thereby increasing the robustness and reliability of the device and dramatically reducing the maintenance of the hardware and extending its useful life-: a) a layer of specific polymers (non-polar compounds) coated on the surface of the optical window (tests during our Phase 1 Feasibility Study revealed positive results[5]) for preventing fouling. Thin PTFE layers were sufficiently transparent in the SWIR wavelengths and reduced the probability of fouling as PTFE is a chemical compound with very low polarity; b) an online sampler concept (validated in the Phase 1 Feasibility Study for grape juice, which is a very difficult liquid product/sample) as an alternative solution to the inline approach for use with difficult aqueous samples prone to foul optical windows: a 2-stages tangential filter -which provides continuous operation-, a CIP (cleaning-in-place) circuit and an ad hoc thermostatic photometer that operates at a challenging wavelength (280 nm) was developed and will be replaced by a NIR spectrophotometer during this Phase 2 project (note: the shorter wavelength, the more difficult to deal with turbidity, fouling and other artefacts, as such if the system is capable of work at 280 nm, it will at work at NIR).

[1] Pemen, A.J.M. et. al. 1998. On-line detection of partial discharges in stator windings of large turbine generators. IEE colloquium on discharges in large machines, 3/1–3/4.
[2] van den Berg, F., et al., Process Analytical Technology in the food industry, Trends in Food Science & Technology (2012), doi:10.1016/j.tifs.2012.04.007
[3] Haibo Huang et. al. (2008). Near infrared spectroscopy for on/in-line monitoring of quality in foods and beverages: A review. Journal of Food Engineering 01/2008; DOI: 10.1016/j.jfoodeng.2007.12.022
[4] Osborne, B.G., 2000. Near infrared spectroscopy in food analysis. BRI Australia Ltd, North Ryde, Australia. Copyright 2000 Wiley, New York, pp. 1–14 (Chapter 1).
[5] As reported in the Technical Feasibility Report that was submitted at the end of the VISUM SME Instrument Phase 1 project (Grant Agreement number: 684767).

Our VISUM platform will mainstream the industry adoption of IR-based reliable and robust on/inline monitoring systems that are readily tailored to the specific measurement requirements and environments and suited to the real-time measurement of physical and chemical parameters in a multitude of food and pharmaceutical applications. The affordability and easy of set up of the VISUM in-line monitoring systems will also open up the power of IR technology to the SME sector.

OBJECTIVES AND EXPECTED OUTCOME OF THE INNOVATION BUSINESS PROJECT

From a business perspective, VISUM Ltd., through VISUM®, is gearing up for a high growth business opportunity. VISUM® will be a key product under the IRIS brand. Over the last 4 years IRIS has positioned itself as an emerging player in the advanced monitoring and control space, whereby we have been making progress in offering the customization of real-time monitoring solutions and the associated integration services on an ad hoc basis to clients in the food, and more recently the pharmaceutical, industry. The expected outcome from this project is to have the VISUM® hardware, software and cloud infrastructure developed to product level by the end of Phase 2, to have the manufacturing set up with the first product units manufactured and quality approved, to have finalized our pricing structure and product positioning. The Phase 2 project enables us to accelerate this work so that we can be ready for market launch in 2019; this acceleration allows us to seize the market opportunity as early as possible and to be first to market with **the world's first IR spectroscopy integration platform**. In parallel to the product development itself, we will gear up and prepare for the commercialization phase, whereby we will build and train our Sales team and we will also establish an Application Team (a technical team that will support sales and deployment activities: testing materials, performing detailed site visits, installing the systems, provide training and handling both site and remote customer support, etc.). We establish our first-tier international distribution network to cover the following geographical markers: Ireland & UK, Spain, Germany and Central Europe, the US and China. We will verify and plan for the realization of our sales objectives- 1300 units being sold (in Europe and worldwide) during the first 5 years of commercialization (rational provided in Section 2.2), which will translate into a cumulative turnover of over €42 M (generated from 2019 to 2023) and will enable 39 new jobs to be created in VISUM Ltd. over that same period.

1.2 RELATION TO THE WORK PROGRAMME

Phase 2 VISUM project relates to the agri-food and bio-based sectors of the ***SMEInst-07-2016-2017*** work programme topic ***'Stimulating the innovation potential of SMEs for sustainable and competitive agriculture, forestry, agri-food and bio-based sectors'***.

1.3 CONCEPT AND APPROACH

CURRENT STAGE OF DEVELOPMENT OF THE BUSINESS INNOVATION PROJECT

A VISUM hardware feasibility prototype device was built consisting of a diffuse transreflectance arrangement associated to a SWIR MEMS-based spectrophotometer. This feasibility prototype has been successfully used in a pilot study with a local company for the determination of moisture (dispersed water) in olive oil and oil concentration in olive marc (alperujo), whereby chemometric models for fine tuning the spectroscopic readings have been developed on the basis of the popular PLS method. The metrological capabilities of the prototype were validated in terms of precision of the predictions: Prediction errors below 0.2% w/w demonstrated that the system offers performance comparable with equivalent more expensive commercial systems. These tests were carried out in order to confirm the applicability of the VISUM concept for online real-time monitoring of an industrial centrifuge (decanter) for yield optimization (PAT). In the Phase 1 SME Instrument **additional trials were performed** with two providers of spectrophotometers, RADIANTIS (900-1700 nm) and Spectral Engines (1550-1950 nm), spectrophotometers in terms of generic performance as well as the adequacy of the prototype for measuring key parameters in palm oil for pastry manufacturing, as well as in meat, olive oil, milk and grape must. Key features such as signal-to-noise ratio (S/N) at relevant wavelengths, spectral resolution, short-term and mid-term spectra stability –in both signal and wavelength-, mechanical and optomechanical stability, and accuracy of the predicted results with representative samples were determined for both spectrophotometers. In terms of key features both devices behaved in a similar way even though the one from Spectral Engines showed slightly better both S/N ratio and spectral resolution, while its spectral range was definitively shorter. In the case of oil for example, the transreflectance geometry was confirmed as the most suitable for measuring semi-transparent liquids, whereby the spectra of 100 different samples from a real process were acquired with VISUM (based on the RADIANTIS SWIR device) and the respective reference analysis were carried out to build quantitative and qualitative (classificatory) models. Predictive models were successfully validated for the following parameters: Total Acidity, Polar-compounds Concentration and identification of degraded (overheated) oil. The Radiantis device results to be more suitable for generic applications –that require wider spectral range- while Spectral Engines spectrophotometer is more suitable for specific applications –that allow narrower spectral ranges and perhaps better S/N- once the respective ideal spectral region has been identified.

FIGURE 2- IMAGE OF VISUM PROTOTYPE HARDWARE AND USER INTERFACE

In relation to solutions towards **addressing the challenge of the biofouling or contamination of the optical window**, which is a major challenge for in-line monitoring in food and pharmaceutical applications, during the Phase 1 SME Instrument project we performed a technical feasibility study into the suitability of three approaches:

1. **Nanotexturisation coating:** A structure of nanometric "pillars" keep the fluids and the solid bodies (microscopic and macroscopic particles) far enough from the surface to avoid stable Van de Waals bonds. In this way, the chance of persistent adherence (fouling) would significantly decrease. Nanotexturisation treatment is under development in the Dr. Pruneri's Group at the Institute of Photonics Science (ICFO) (Dow Corning Cathedra) and they provided treated and control samples for a feasibility study. Results with the current nanotexturised windows performed on water, milk and powder (flour), revealed positive results for reducing fouling in the case of certain powders such as flour, but currently would not be suitable for use for reducing fouling in the case of water-based solutions or lipid-based emulsions (such as milk). ICFO is still working with the technology and we will work together in the future in order to find a robust solution for the industry. However, due to insufficient TRLs this work will not form part of the Phase 2 SME Instrument. The results are summarised in Table 1.
2. **Antifouling polymer layer coating:** A layer of specific polymers (non-polar compounds) is coated on the surface to reduce the probability of fouling. In particular, a thin PTFE layer would be transparent enough in the SWIR wavelengths and would reduce the probability of fouling because PTFE is a chemical compound with very low polarity. The results of feasibility tests performed in the Phase 1 project, a Dupont AF Teflon disk (approx. 200 microns thick) was placed on a 1 inch diameter 2-mm thick sapphire optical window and held on it by means of a mechanical holder. The transmittance along a wide spectrum (400 to 2000 nm) resulted over 93% -which is perfectly suitable for optical applications such as VISUM inline windows. PTFE is an almost perfect non-polar substance with excellent antifouling properties and chemically is very inert in relation to the most of the chemicals. Likewise, PTFE is very stable in a wide range of temperatures (<250 C). The sapphire substrate is needed for providing mechanical robustness to the PTFE film; the Teflon layer is a barrier between the analysed product and the sapphire window. As in the nanotexturised window feasibility study, the PTFE-AF-coated window was tested against water, milk and powder (flour). The tests were carried out according to the same scheme that was used with the nanotexturised widows, i.e., pouring the tests substance on the PTFE layer and flipping the set up vertically after 5 min. The results were significantly better because no persistent fouling took place even without any cleaning process (refer to Table 1). In conclusion the PTFE film on a sapphire glass presents a suitable solution for addressing the majority of the cases of VISUM Inline applications). During the Phase 2, we will develop a coupling solution for holding the thin polymer layer in touch with the glass (sapphire) window if required.

TABLE 1- SUMMARY TABLE OF THE RESULTS OF ANTIFOULING COATING FEASIBILITY STUDIES

Sample	Test	Nanostructured layers	PTFE-AF coated sapphire	Control (bare sapphire)
Water	Discrete drops	+	+	-
	20 ml water flow	+	+	-
	Immersed in water during 3 h	+	+	-
	Pouring water drops after 2 days (immersed during 3 h and then dried)	-	+	-
Milk	Discrete drops	-	+	-
Corn flour	Powder layer	-	+	-

3. **Online sampler:** The option of on-line sampling has been also considered. In many cases, inline real-time measurements are not absolutely necessary; on the contrary, in such cases, a sample can be automatically taken from the pipe in order to analyse it by means of an external online setup, properly attached to the pipe. It allows using ad hoc cleaning systems (CIP) that remove any fouling from the windows without interacting with the product within the pipe. To study the feasibility of such an approach, an automated online sampler for liquid samples was designed, built and successfully tested by means of an especially challenging or difficult liquid product/sample: grape must. In fact, grape juice can be considered an extreme case due to its high sugar content (>200 g/L), very high turbidity (>4000 NTU), protein-based colloids content- which easily causes clogging in almost any filtering system- and last but not least, its proclivity to be easily transformed by microorganisms such as yeast -which implies high CO2 concentration (bubbles occurrence) and biofouling too. Such a kind of sample represent a worst case scenario for food industry applications because

combines several drawbacks associated to the samples that are typical in such a sector. The sampler was provided with a 2-stages tangential filter -which provides continuous operation-, a CIP (cleaning-in-place) circuit and an ad hoc thermostatic photometer which operates at a difficult wavelength: 280 nm. Such a short wavelength is particularly sensitive to turbidity and any kind of fouling hence it resembles the worst case scenario. The photos above (left) show the online sampler lab system and (right) the sampler installed in the lab in an ad hoc test rig for simulating real conditions: the grape must be continuously pumped through a closed-loop pipe and samples are continuously taken out by means of the sampler to be properly conditioned (filtered) and characterized by means an ad hoc photometer. A summary of the parameters that have been obtained are shown in Table 2. We conclude that the online sampler concept implemented in the test rig is suitable as an alternative solution to the inline approach for difficult aqueous samples prone to foul optical windows and will be implemented in Phase 2.

TABLE 2- SUMMARY OF THE PARAMETERS OBTAINED WITH THE SAMPLER UNIT

Turbidity after the filtration procedure (starting > 2000 NTU)	< 10 NTU
Repeatability of optical absorbance @ 280 (2σ)	< 1%
Warm-up time of the system	80 min
Delay of the measurement	1 min.
Equilibrium temperature of the photometer housing	24.5 ± 0.5 ºC

From a **market feasibility perspective**, we performed significant industry engagement prior to and during the VISUM Phase 1 project, whereby we participated in the following trade exhibitions: Murcia Food Brokerage 2015 (14-15 May), BTA Barcelona Food Technologies (May 21-24 2015), Global Food Safety 2015 in Amsterdam (September 8-9, 2015), Anuga Food Congress in Cologne (October 10-14 2015), and PharmaProcess in Barcelona (October 27-29, 2015), III International Food Safety Congress, Murcia (25- 27 Nov 2015), IFPAC (Conference on Process Analytical Technology (PAT), QbD, Process Understanding & Control, and Real-Time Analytics) US (Jan 24-27 2016). Extensive industry engagement has confirmed very strong industry interest in the VISUM platform and its value proposition. We have generated 106 new leads in our CRM (79 from the food industry and 27 from pharma) and from them we have recruited four companies to collaborate with us in the Phase 2 as 'first beta clients': SALGOT (meat processing); DAIRYGOLDPASCUAL (dairy processing); METALQUIMIA (olive oil production); and d) ALMIRALL (pharmaceuticals).

During the Phase 1, face-to-face meetings were held with the R&D+I, Production and Quality Managers at each site in order to a) become familiar with their facilities and processes to be monitored, b) define the user requirements (using a Kano Survey), c) implement the Quality Function Deployment (QFD) method to determine how and where priorities identified through the Kano Survey are to be assigned in product development, d) define the derived system specifications for the VISUM system, which is the starting point of this Phase 2 project for designing and manufacturing a mainstream VISUM platform that will be flexible enough to ensure rapid customization for tailor-made VISUM solutions. These user requirements, which have fed into the system specifications (outputs from Phase 1) are summarised in Table 6 in Section 2.1.

POSITIONING OF THE BUSINESS INNOVATION PROJECT

In the Phase 2 we will advance the TRLs from the current TRL6/7, whereby the VISUM prototype has already been successfully demonstrated in a relevant environment, to hardening up the hardware and building the software to perform validation and demonstration work in the industry (TRL7) and arriving at a complete and qualified system, which will be CE marked (TRL8), and through continued case studies in the industry, the fully industrialised systems will be proven in an operational environment (TRL9). We will also get to Manufacturing Readiness Level 10 in order to be in a position to serve market demand when we launch in January 2019.

TABLE 3- SUMMARY OF THE TRL ADVANCEMENTS THROUGH THE PHASE 2

Key element	Current TRL	Final TRL	Comments
NIR Spectrophotometer	8/9	9	VISUM is based on of-the-shelf spectrophotometers that may involve some minor OEM adaptations
Antifouling window	6	8/9	Optimized and tested for the most typical types of compatible samples
Lighting	6	9	Amelioration in terms of geometry (variable beam diameter), filtered spectra –to reduce sample heating- as well as pre-heating the lamps to reduce warm up time and improving stability
Software	6	8/9	Improvements in terms of usability and robustness as well as regulatory compliance (11 CFR 21)
Casing	7	9	Optimization of size, thermal stability and regulatory compliance (ATEX, CE, etc.)
Data exchange interface	7	9	Compatibility with the most common standards (Modbus, Profibus, Fieldbus, etc.)
"miniPAT" control	6	9	Completing a set of basic control routines and methods (PID, MSPC, etc.)
Online sampler	6	9	Optimized and tested for the most typical types of compatible samples
Cloud database	8	9	Communication between the VISUM devices and cloud will be verified.

THE VISUM CONCEPT

NIR is a non-destructive, reliable and accurate technique for monitoring chemical and physical parameters during food and pharmaceutical processing. At present, NIR is widely accepted as one of the most promising on/in-line process control techniques. Moreover, the appearance of fibre optic probes significantly improves the ability of NIR techniques to monitor and control processes especially using remote on/in line detection. However, as was seen, there are currently gaps preventing the mainstreaming of NIR in the industry: 1) typical NIR lab analysers are expensive (~40-80K€) and they require specialised staff to operate them, along with costly maintenance; 2) they often require time-consuming, costly and complex predictive models development and their further recalibration

(model maintenance); 3) Generic systems do not properly fit to each particular product line or monitoring challenge and these mismatches are the source of major errors; 4) on-line NIR optical interfaces are prone to biofouling and contamination which is a major limitation for accurate measurements and leads to high maintenance requirements.

The VISUM platform (the architecture layers are outlined in Table 4 below) is designed to overcome each of these limitations and thereby contribute to the mainstreaming of NIR-based technology in the European and global food and pharmaceutical industries as an enabler for improved product quality and safety, as well as improved process understanding and control for a first tier Process Analytical Technology (PAT) tool. By ensuring that the VISUM platform can be customised to real-time industrial solutions at a cost to the client of on average ~€10-35k depending on the final application, we will enable **a totally disruptive costing structure that will mainstream the adoption of NIR-based technology in the European and global food and pharma industries, making it fully accessible to SMEs.** Moreover, as was discussed we will ensure VISUM is enabled by cloud computing in line with the Internet-of-Things and the Industry 4.0 paradigm.

TABLE 4- ARCHITECTURE OF THE CLOUD-ENABLED VISUM PLATFORM

VISUM HARDWARE LAYERS	
VH.1	**Tailor-made probe-sample interface ('design for operation')**
A set of basic arrangements aimed at the seamless adaptation (integration) to the most common/standard user situations, as well as to specific needs in terms of sampling (probe-sample interaction). Generic systems do not properly fit to each particular product line or monitoring challenge and these mismatches are the source of major errors. To overcome problems associated with biofouling and contamination of the optical window from powders or viscous liquids, an antifouling polymer layer coating technology feature suited for most inline applications, and an online sampler unit with CIP features for other applications, will provide practical and affordable solutions for preventing fouling and increasing the reliability of in-line solutions, as well as dramatically reducing system maintenance and extending performance life.	
VH.2	**Core instrument: IR spectrophotometer with antifouling nanotexturised coating**
The standard VISUM core device is a MEMS-based spectrophotometer that works in the SWIR range (0.9-1.7 µm). MEMS technology guarantees not only affordability [unit price can now be as low as 1 k€ depending on requirements] but also robustness –in terms of being suitable for applications in harsh environments- as well as reduced both size [volume < 100 mL] and power consumption [<10 W]. This core device shows good signal/noise ratio [better than 2500:1], which is sufficient for the majority of the typical applications. In addition, the platform is also highly scalable and offers the possibility for a "by-design" option enabling the use of additional devices depending in the specialised requirements of the end-user and also for ensuring the VISUM platform can be seamlessly updated and upgraded in line with the evolution of MEMSNIR technology.	
VH.3	**Built-in data management hardware**
Miniaturized embedded computer	
VISUM SOFTWARE LAYERS	
VS.0	**Cloud database and Application Programming Interfaces (APIs)**
The secure cloud infrastructure will allow the data from the VISUM devices to be sent to the cloud (optional service) and accessible over a web application (for PC and mobile interfacing) to enable remote access, and whereby the historical data can be mined using data analytics tools.	
VS.1	**Built-in data management software**
For driving the core instrument, data processing [chemometrics modelling and prediction] communication and the built-in interface, when necessary. The proprietary software will be compatible with popular chemometric packages –e.g. SOLO [Eigenvector] and Unscrambler [CAMO]- in terms of easy integration of predictive models generated by them if necessary. It allows users to independently build and maintain the chemometric models. For pharma applications the system is pre-validated according to the regulations (e.g. 11 CFR 21).	
VS.2	**Ad-hoc chemometrics modelling**
While the models can be developed by the Client by using high-end third-party software, the platform is intended for working on the basis of automated tailor-made robust multivariate calibration models in order to guarantee the maximum reliability and accuracy as well as reducing the cost for the end user. Automated assisted modelling solves the main drawback of applying chemometrics to process control: the intrinsic significant cost related to the predictive models development. The system includes the latest model transfer procedures.	
VS.3	**Flexible multiplatform data interface**
The result outputs (analytical data) can be sent to the Client's data management system through a variety of methods (wifi, standard databuses, etc.)	
VS.4	**User programmable decision rules**
This offers "mini-PAT" built-in capability, whereby the user (client) is allowed to code ("scripting"/macro) its own decision rules in order to provide the decision system with "action-type" data. Close-loop control can be easily implemented by the industry end users without incurring extra costs and according their production criteria through standard protocols like MODBUS or TCP/IP which easily interface with the facilities' control system (SCADA).	

APPROACH/ACTIVITIES THAT WILL BE IMPLEMENTED DURING THE PROJECT

Our VISUM project follows an **application-oriented product development approach**, which will involve a tight co-creation between the development team and our target client base. A number of first clients have agreed to participate in the project as early adopters to permit the project to either, on the one hand, use real product samples as a basis for the development, and/or provide access to their facilities for industrial trials (a summary of the target applications is provided in Table 5).

TABLE 5- TARGET APPLICATIONS FOR THE PHASE 2

Company	Sector	Target Products	Participation Level	Parameters to be measured	Installation

ALMIRALL	Pharmaceuticals	SEKISAN (liquid)	Provide samples for model building	Target compounds: Cloperastine fendizoate [3.5 mg/mL]; Nipagin [1.22 mg/mL]; Nipasol [0.18 mg/mL]	Blender Dosage filling machine
		ACECLOFENAC (solid form)	Provide samples for model building	- Dissolution test - Moisture - Hardness and degradation by-products - Target compounds: Aceclofenac [488 mg/g]	
		ALGIDOL (solid form	Access to facilities for industry trials	- Blending end-point - Moisture - APIs content - Target compounds: paracetamol [300 mg/g], codeine [4.5 mg/g], ascorbic acid [227 mg/g]	
CALIDAD PASCUAL	Dairy	Smoothies (liquid)	Provide samples for model building	- Mixing end-point	In-pipe
		Fruit juices (liquid)	Provide samples for model building	- pH, glucose, fructose and total acidity	
		Milk (liquid)	Provide samples for model building	- Water content [expressed as cryoscopic point from 0.510 to 0.530 mK, precision at 0.002 mK] - Fat [0 to 7.4 %w/w] - Proteins [2 to 4 %w/w]	
ACESUR	Olive oil (and olive-oil-based products)	Sauces (liquid)	Provide samples for model building	- Brix and end-point of CIP process	In-pipe (in the bottling line)
		Mayonnaise (semi-liquid)	Provide samples for model building	- Fat content [65% w/w]	
		Olive oil (liquid)	Provide samples for model building	- Fat content - Moisture [0.04 to 0.08 %w/w, precision at 0.01% w/w]	
SALGOT	Meat	Salamis and sausages filling mixture	Access to facilities for industry trials	- Moisture - Fat content - Protein content	At the output of the stuffer line
DAIRYGOLD	Dairy	Milk	Access to facilities for industry trials	- Fat [0 to 7.4 %w/w] - Proteins [2 to 4 %w/w]	In=-pipe
METALQUIMIA	Food Processing	Snack Food	Access to facilities for industry trials	-	Conveyor oven
DEC	Pharmaceuticals	APIs(Confidential)	Access to facilities for industry trials	- Blending end-point - Moisture - APIs content	Blender in-pipe

Using the feasibility prototype system that is available in VISUM we will undertake an initial demonstration of the system on the applications to provide test data for the development project. An iterative product development cycle will be implemented to design and produce beta prototype systems that will be installed at each of our end-user sites as test cases for performance verification, the success of which will confirm the acceptance of the prototype designs. Having validated these designs, we will **translate the prototypes to product** through the creation of the manufacturing documentation package using the engineering information and specifications and we will set up our manufacturing facilities and produce and quality approve the first units. In relation to methodologies that will be used during the development work, our teams operate under agile frameworks, whereby Kanban is employed for hardware developments, and Scrum and Test Driven Development (TDD) for software developments. Whenever creative thinking is needed, particularly for ideating specific solutions that cannot be addressed by finding a simple trade-off among several technical constraints, the development will be based on TRIZ (Theory of the resolution of invention-related tasks), which is a widely recognized methodology for accelerating and simplifying the search for inventive and innovative solutions and a scientifically founded approach to the forecasting of the evolution of technological systems, products and processes[6]. The VISUM TRIZ Team has been properly trained by recognized members of ETRIA (European TRIZ Association[7]) for undertaking creative challenges associated to R&D tasks. The TRIZ approach will be particularly convenient for finding flexible (versatile) designs compatible with the most of the applications to be addressed by VISUM.

[6] TRIZ. Innovation and Inventive Problem Solving Handbook", Pavel Livotov and Vladimir Petrov ©TriS Europe, Berlin 2014. www.tris-europe.com
[7] http://etria.eu/portal/

In parallel, we will implement **market development activities,** whereby we will further develop our business plan (building from a Business Model Canvas) and, based on planning within the framework of the classical and proven 4 Ps (Product, Pricing, Promotion and Place) Marketing Mix, we will perform comprehensive activities to gear up for the Commercialisation phase.

HOW THE PROJECT CONCEPT AND OBJECTIVES FIT INTO THE OVERALL PLAN TO REACH THE MARKET

This SME Instrument project is fundamental for realising our overall plan for reaching the market by 2019. VISUM® is a development that requires acceleration. To date, through own internal R&D funds, we have invested ~30 months and ~€220K of our internal resources in getting to TRL6/7. The ability to leverage public funds to complement our private investment will provide the acceleration platform that is needed to complete the development and bridge the valley of death in going from TRL6/7-TRL9 and into the market. Through the SME Phase 1 Instrument we have had the opportunity to perform a feasibility study in relation to tackling the biofouling problem and we have defined the most appropriate solution to be developed, we have also gained greater market insights, formulated a clear business proposition and identified our market positioning. We have also recruited 4 beta testing end-user companies and together with them we have researched their requirements and defined the system specifications. The design, engineering, model development, validation and demonstration work that will be performed in the Phase 2 Project in order to test and verify the reliability, repeatability, and robustness of the system hardware, software and models, as well as the operational and industrial-level performance of the VISUM® development, in addition to the certification, manufacturing and commercialisation planning/market development work that will be performed, are all important steps of a product development on its way towards commercialisation. With an injection of funds, and by gearing up right from system specifications, through design, prototyping, industry trials in compliance with all industry standards and regulatory requirements, we will ensure that there are no user-, technical-, market- or regulatory- obstacles to the uptake of the technology in the marketplace.

HOW THE PROJECT DEVELOPS SOMETHING NEW TO EUROPE AND ADDRESSES EU-WIDE CHALLENGES

VISUM® is the first flexible IR integration platform of its kind that will enable the affordable customisation of PAT-oriented NIR-based real-time monitoring and quality control solutions targeted at the European and global food and pharmaceutical industries (and sufficiently affordable and easy to use and maintain to open up the SME sector). The ability to leverage the SME Instrument to accelerate the transformation of new knowledge for industry to conceive and develop new products, industrial processes and services, is an important tool for creating a true Innovation Union: a European Union that thanks to its research and innovation performance can support sustainable growth.

From the **technology side**, photonics is regarded as a driver of innovation and plays an important "enabler" role for many industrial and customer markets. With a worldwide market value projected to hit €615 billion in 2020 and to remain stronger than the gross domestic product (GDP), photonics has moved from a niche application to a key enabling technology, according to Photonics Industry Report 2013. The European photonics industry holds about 55 % of the world market in production technology. VISUM ® is an enabling platform that will significantly increase spectroscopy adoption in industry, and in doing so it will leverage and strengthen existing European expertise in MEMs-IR spectroscopy (from European sensor providers, such as Spectral Engines from Finland and Radiantis from Spain) which will help Europe to retain its leading position at the front of the field, whereby we are currently ahead of Asia or North America in sensing and spectroscopy. VISUM will contribute to European leadership in the sensing revolution by bringing a 'Made-in-Europe' solution that combines all the existing technology platforms (such as affordable MEMs sensors, chemometrics, IoT and cloud computing) to further increase EU leadership in the photonics field.

From the perspective of the **end-user industries**, increasing emphasis on product quality and safety, and on production processes and process efficiency- and on the relationship between the two- has fuelled research and industry interest in the development of on-line detection techniques that enable detection of quality changes of raw materials and final product under steady process conditions[3], as well as fraud and counterfeit early detection. Indeed, Process Analytical Technologies are increasingly being used in the Pharmaceutical industry, along with an increasing number of process industries, such as Food and Chemical Processing, as enablers for in-process monitoring and control, providing real time information to the engineer and reducing the need for off-line laboratory Quality Assurance (QA) facilities. They also provide invaluable process data for increasing process understanding and knowledge, and are an enabler for the transition from batch to continuous processing, whereby real-time monitoring is required. The potential for VISUM to advance the European process industries towards continuous, more flexible processing to respond to market trends for customized products as well as bringing process industries into the smart manufacturing Industry 4.0 revolution would have high impacts for European industrial competitiveness. This is in line with H2020 goals for an Innovation Union, whereby the EC is working hard to stimulate job creation and prosperity as an exit from a severe economic and financial crisis. VISUM® will serve as a flagship project for contributing to these goals and for the generation of much needed wealth, growth and jobs, which is Horizon 2020's ambitions[8].

HOW GENDER ANALYSIS IS TAKEN INTO ACCOUNT IN THE PROJECT'S CONTENT

VISUM® product development will consider gender related design features to ensure the needs of female engineers and line operators are analysed and any gender singularities are embedded in the design. During the co-creation process that was implemented in Phase

[8] Brussels, 30.11.2011. COM(2011) 808 final. Horizon 2020 - The Framework Programme for Research and Innovation

1, female engineers/developers and female business professionals from the technology, food and pharma industry were involved and their opinions and contributions gathered. Moreover, in relation to business behaviour and decision making, gender analysis will also be important, especially in view of growing trends and commitments to see more women occupy high level and decision-making positions in industry. The Project Manager for the project is a female Marie Callanan) as are numerous other members of the team (see section 4.1). With regard to the employment opportunities that will be generated in VISUM for the implementation of the innovation project, and subsequent jobs through the commercialisation of the project outputs, a recruitment policy will be in place designed to ensure a reasonable level of balance between the sexes.

1.4 AMBITION

NOVELTY OF THE INNOVATION BUSINESS PROJECT

Optical techniques using **near-infrared spectroscopy (NIRS)** have received considerable attention as a means for non-destructive sensing of food and pharmaceutical quality and safety. Compared to other non-destructive techniques, NIRS spectroscopy does not need any sample preparation, allowing for simple and rapid analysis, which is a requirement for on-line application[3]. It can **simultaneously measure several constituents or multiple quality attributes**. Moreover, the relatively weak absorption due to water enables high-moisture products to be analyzed[4]. All these properties make the NIRS technique widely accepted as one of most promising on/in line detection methods in food, pharmaceuticals and other areas. By plugging the 'integration gap', whereby we shift the emphasis from NIR device to fully integrated NIR-based in-online process monitoring solution tailored to the process, our ambition is to mainstream the uptake of NIR in process industries of all sizes (including SMEs, and one of our first beta clients is a meat producing SME, SALGOT), in order to enable the implementation of real-time process monitoring and control for bringing the process industries into the Industry 4.0 revolution and knowledge based process manufacturing.

An underlying uniqueness in the VISUM platform for the European and global food and pharmaceutical industries is that it is based on "solution" vs. "device/instrument", which will resonate well with production and quality managers. This is a fundamental shift in business approach, whereby having a platform in place that is readily adaptable and customisable to the processing realities of each particular industry user scenario will break the mould in terms of monitoring systems, in that the current market offering requires the user to adapt their internal scenario to the constraints of the device. Some of the 'out of the box' solutions that are coming on the scene in recent years pose problems when the samples to be predicted are measured on a different instrument or under differing environmental factors from those used to build the prediction models, whereby the changes in spectral variations between the two conditions can make the model invalid for prediction in the new system[9], requiring a complete recalibration, which is time-consuming and costly. Oftentimes this service is not provided by the device provider, who specialises in the commercialisation of instruments as opposed to solutions, providing an additional barrier for end-user industries. VISUM overcomes one of the greatest challenges/risks preventing the adoption of in-line monitoring systems in the industry, and in particular in the SME community- that of **affordable integration**. In doing so we open up a major opportunity to tap into a SME market that has currently been left unattended by the larger device manufacturers.

EXPECTED MARKET APPLICATION THAT DIFFERENTIATES THE PROJECT AND PROVIDES THE HIGHEST ADDED VALUE

In relation to the key market applications, we are targeting the **food and pharmaceutical process industries** initially (and Table 4 summarises the first applications that the VISUM will be validated for), and our intention is to target other process industry (chemical processing, ceramics, minerals…) and agriculture applications in the post-project phase. To open up these applications, VISUM will remove the high doorstep that traditionally prevents the adoption of NIR technology:

- **high equipment costs** (which will be overcome by the implementation of very low-cost MEMS-NIR core sensor);
- **complex, time-consuming and costly re-calibrations** (overcome by the VISUM calibration transfer models);
- **challenges posed by differing and oftentimes hostile or tricky operating environments** (overcome by our tailor-made probe-sample interface for optimal probe-sample interaction, to avoid mismatches that can lead to major errors)
- **fouling and device maintenance** (overcome by our novel antifouling polymer layer coating technology or online sampler unit with CIP features depending on the application)
- **integration** (which requires specialised knowledge in spectroscopy and chemometrics for ensuring reliable operation in/on-line and overcome by our 'integrated-by-design' approach to the platform that will ensure integration can be performed by engineers that do not require a specialisation in spectroscopy)

In order to open up the tremendous benefits of NIR-based in/on-line monitoring technology to mainstream applications in the food and pharmaceutical industry, and especially for the SME sector, VISUM is hinged on a number of key advancements:

1. The **integration of the latest developments in affordable** (e.g. Spectral Engines now offers a price point of just under €1k for orders of 10 sensors)**, compact and reliable MEMSNIR sensors** with no macroscopic moving parts, signal/noise ratio better than

[9] Robert N. Feudale, Nathaniel A. Woody, Huwei Tan, Anthony J. Myles, Steven D. Brown, Joan Ferre (2002). Transfer of multivariate calibration models: a review. Chemometrics and Intelligent Laboratory Systems 64 (2002) 181 – 192

5000:1 in the most significant spectral region, spectral range which covers SWIR (from 0.9 to 1.7 μm), minimum size (a few centimetres) and low power consumption (<4 W).

2. The tools for easily achieving **better multivariate calibration models** will comprise the latest advances readily implemented in a software suite that will maximize the model quality in a totally automated environment. Chemometrics will play a major role in giving accurate results, by acting as a "noise rejection" filter, where the "noise" here is the sum total of all the extraneous effects that act on the instrument, and through it, on the data. The implementation of the newest chemometric algorithms and tools in the VISUM software -like efficient variable selection by Recursive Weighted Partial Least Squares (rPLS), the Tikhonov Regularization for ill-conditioned data as well as model transfer (standardization) by Stacked Partial Least Squares (SPLS) regression- will address more robust predictive capabilities without increasing time, efforts and related cost. It is widely known that providing the right training dataset is the most relevant cost source in IR spectroscopic-based monitoring. More powerful and efficient chemometric techniques would dramatically simplify the need of collecting a huge number of spectra and their related reference values by traditional methods. The end user will be **capable** of automatically building excellent predictive models without requiring any expertise in Chemometrics. Particularly, the Platform will comprise: a) Automated routines for selecting the most suitable method among the best ones (PCR, PLS, SVM, ANN, etc.) for each "training" dataset; b) Intuitive "wizards" for guiding the user along the modelling and standardization (model transfer) processes; c) Automated routines for outlier detection and rejection; d) User access to define the automation settings and criteria –other than the default ones
3. **Anti-fouling solutions** based on: a) a novel layer of Thin PTFE specific polymers (non-polar compounds) coated on the surface of the optical window; b) an online sampler concept for difficult aqueous samples. These solutions would represent a step change in overcoming the fouling and contamination of the optical window, resulting in dramatically lower (~<80%) maintenance requirements and replacements, as required with traditional optical windows.
4. **Ready-to-install capabilities**: The VISUM Platform is provided with the adaptations for the most frequent sampling cases –readily implemented in both hardware and software. This will result in a dramatically reduced cost (~up to 4-5 times cheaper than existing commercial systems) for a fully integrated solution, which could be performed by non-spectroscopy specialized engineers.

EXPECTED PERFORMANCE/IMPACT ON DEFINED NEEDS

Companies are under increasing pressure to ensure product quality and safety and increase efficiencies. There is a big market opportunity for affordable, robust and reliable real-time monitoring technology to assist EU and global food, pharma and other manufacturers to respond. When VISUM is deployed in industry and in use, it will enable reliable real-time process control based on on/inline monitoring of the Critical Control Parameters (CCPs) and the Critical Quality Attributes (CQAs). The impacts for end-users against which a Return on Investment will be measured include (depending on the application): ■ability to control the process and reliably determine end-point (which will avoid extending processing time and reducing or eliminating traditional QC laboratory analysis which stop and delay the production cycle); ■ reduced dependence on laboratory off-line and post-production analysis that are time consuming and require specialized staff; ■ standardized (or increased) product quality; ■ increased batch yields; ■ reduced process down-time (due to processing problems); ■ reduced incidences of process deviations and off-spec product (leading to reduced discard and minimized risk of supply of off-spec to the market); ■ greater assurance for regulatory bodies/market; ■ increased process understanding and the ability to implement Quality-by-Design to increased process efficiencies.

As is discussed in Section 2 (and summarized in Table 7), the current commercial offering of leading NIR providers such as Foss, Thermo and others, centres on the sale of devices suited to solving universal monitoring needs that respond to a wide market and cannot be easily tailored to specific challenges faced by individual processors, as they lack the necessary robustness in relation to the calibration models. Moreover, their business models are centred as instrument providers, and not as tailored solutions providers- the vast majority of the instrument providers in the market do not offer integrated solutions, i.e. they do not perform integration work, nor do they develop or validate the chemometric models, as their core business the sale of instrumentation. This places an important and costly challenge on the end-user industry to get the device operating reliably in their line; posing a prohibitive barrier for many SMEs. Their integration services in a PAT context are aimed at large projects related to huge facilities or extreme added value production lines. Consequently, an important market segment –industrial SMEs needing affordable turnkey solutions- cannot access such technologies. Furthermore, the biggest players have been slow to upgrade their technologies and adapt their pricing structure to the new context; most probably given that a 'business as usual' scenario serving the higher end, more costly applications is very profitable. That said, they are beginning to take note, and FOSS, for example, has recently launched the ProFoss in-line NIR system for dairy applications. However, end-users need an off-line FOSS device to calibrate the in-line system on sampling, which means that it is more than likely existing FOSS customers who will upgrade to the new in-line system.

In conclusion, the main advantages of our VISUM® solution with respect to competing solutions is that it provides 'by-design' a unique solution that combines a complete set of PAT functionalities in a "plug & play" probe with an affordable price. However, as every product is subject to a lifecycle including a growth phase followed by an eventual period of decline as the product approaches market saturation, in order to retain the competitiveness of VISUM in the market, we will plan continuous product extensions though innovation. For example, the flexible VISUM platform will be able to evolve in tune with the latest developments in terms of MEMS spectroscopy and advancements

in chemometric knowledge to continually improve. Moreover, our initial product is based on SWIR (1000-1700 nm), which offers an optimum trade-off between price, maturity, penetrability and chemical sensitivity. We will also develop a NIR (800-1000 nm) VISUM product, which will bring advantages for extended applications, whereby for example, due to the relatively weak absorption due to water, NIR enables high-moisture products to be analyzed[4] as well as E-SWIR (1000-2500 nm) for extending the applications to more complex matrixes. Moreover, by ensuring VISUM is cloud enabled, we have the ability to offer data services in relation to the mining and analysis of historically stored data- the adoption the Big Data paradigm in the context of process data- and as a central enabler for Quality-by-Design, greater process understanding and more efficient processes.

2. IMPACT

2.1 EXPECTED IMPACTS

A) USERS / MARKET

USER NEEDS THAT ARE BEING MET

VISUM® is **the world's first IR technology integration platform** for enabling the cost-effective and ready customisation of robust and reliable in-process monitoring solutions in food and pharmaceutical processing applications, among others. It responds to an industry **need for improved process and quality control that requires a move away from time consuming and costly laboratory-based quality measurements and post-production quality control (QC) that do not provide processors with access to information in real-time, thereby delaying action that can lead to product discard, with cost and environmental implications.**

VISUM is a timely and relevant product development, as industry leaders and decision makers- in response to market, regulatory, sustainability and competitive pressures- are starting to take note of the importance of real-time monitoring technology as an enabler for: ■process efficiencies; ■process flexibility; ■ product quality standardisation (the ability to arrive at consistent product quality that meets with market expectations every time); ■reduced dependency on time consuming and off-line analytical measurements; and ■ accurate determination of process end-point to avoid over-processing (which not only leads to higher production costs and increased energy use, but can also result in spoiled or down-valued end-product or even incidences of e.g. lumber (solid materials in pipes and reactor)). All these benefits can have multibillion euro annual impacts for industries. David Scott from DuPont Research[10] sums it up well: "*As manufacturing processes become increasingly complex, industry must rely on advanced sensor technology and process control to improve efficiency and product quality. Processes now need a variety of on-line measurements*". Given the need to achieve better control, measurement of process parameters in real time is seen as vital to process understanding, industry needs access to fast, reliable and accurate tools and methods for providing in-situ measurements[10].

Beyond the holistic needs, and in order to look deeper at the specific application-centred requirements of industry in relation to in/online analytical technology, during the Phase 1 SME Instrument VISUM project, an analysis of the industry requirements was carried out by means of a Kano Survey (which was reported in the Technical Feasibility Report at the end of the Phase 1 and was used to define the system specifications) among the 4 beta clients/early adopters who cooperated with us in the Phase 1 (ACESUR, CALIDAD PASCUAL, SALGOT, ALMIRALL). These and the addition early adopters that we are now engaging with cover a representative variety of liquid, semi-liquid and solid foods and drugs, measurement parameters and integration scenarios (blender, in-pipe, etc.)- refer back to Table 5- and they are also presentative of large, medium and small companies. Kano is part of the Quality Function Deployment (QFD) methodology that we have used to collect and prioritise client expectations and needs without missing any important details. The main benefit of using this methodology is the direct correlation from customer's language to a "black and white" language for developers. A universal requirement- that was confirmed through the Kano Surveys- is the need to have NIR probes that are suitably matched to the particular circumstances of each facility. However, we also identified numerous similarities that can be exploited in terms of adapting a generic solution, which is in line with the spirit of the VISUM, and that will enable us to design, for example, a **universal sampling approach** (based on similar requirements or constraints in terms of speeds and sampling geometry (diffuse reflectance) and common challenges in relation to bubbles, foam, lumps, etc.), **universal chemometrics** (as the nature of the samples- powders, semi-solid and turbid liquids- is similar), data exchange through an **Ethernet cable is compatible with each client** and, likewise, all of them require the same ingress protection (IP) grade, standard coupling formats can be used for the mechanical integration in the production lines ("Triclamp" is mainly used in pharma while NW or DIN is commonly found in the food sector), among others. All these conclusions confirm that there is a very real opportunity to standardize to a significant extent and consequently achieve the main goals of the VISUM platform. In relation to the user needs or requirements that will be met upon completion of the project, the 20 most significant requirements from the KANO Survey are provided in Table 6- each of which will be fulfilled by the VISUM platform and have been contemplated in the system specifications.

TABLE 6- REQUIREMENTS FOR FULFILLING CUSTOMER EXPECTATIONS

[10] Scott, D., M., 2008, Industrial process sensors. CRC Press

No.	Requirement	Type of customer factor	Type of requirement	Value	Importance rating (0-5)
1	Success in classification	Basic	Non-functional performance requirement	> 90 %	3
2	Confidence interval in classification	Basic	Non-functional performance requirement	95%	5
3	Spectral range	Basic	Non-functional performance requirement	900-1700nm	3
4	Measurement period will be minimized	Linear	Non-functional performance requirement	<20 s for on-line sampler; 1 s for inline	4
5	Weight lower than 3 kg (in-pipe attachment) and not relevant for fixing at architectural structures	Linear	Non-functional system requirement	Yes (<3kg)	3
6	Minimum IP (ingress protection)	Linear	Non-functional system requirement	IP65	3
7	Food grade materials in the interface with products	Basic	Non-functional performance requirement	Yes	5
8	Environment temperature	Basic	Non-functional system requirement	-5 to 60 ºC	5
9	System withstands process pressure	Basic	Non-functional system requirement	6 Bar	5
10	System withstands process pH	Basic	Non-functional system requirement	2-12	5
11	Sampling port for aliquots (reference samples for modelling, validation or calibration)	Basic	Non-functional system requirement	Yes	4
12	HMI via software installed in PC	Basic	Non-functional implementation requirement	Yes	4
13	Control software will show instant measurements in graphs together with acceptance limits	Basic	Non-functional performance requirement	Yes	3
14	System will include data log	Basic	Non-functional performance requirement	5 Years	3
15	Embedded –PC wired communications (Ethernet)	Basic	Non-functional implementation requirement	Yes	4
16	Models and prediction machine operate in the built-in computer (probe)	Excitement	Functional requirement	Yes	2
17	Software 21CFR part 11 compliance	Basic	Functional requirement	Yes	5
18	Software designed to allow the user to include more models in the future	Basic	Non-functional performance requirement	Yes	4
19	At least prepared to measure in diffuse reflectance geometry	Basic	Non-functional implementation requirement	Yes	3
20	Wireless data link for blenders	Linear	Non-functional implementation requirement	Yes	4

VISUM, by enabling the deployment of cost-effective and easy to install NIR-based in/on/at line monitoring solutions that through the new chemometric and calibration models offered by the system, will represent a step-change in terms of reduced set-up costs and complexity and will allow for true 'out-of-the-box' NIR solutions that can be installed by engineers with no need for specialized spectroscopic knowledge. This will respond to the needs of small, mid-tier and large manufacturers alike. Closing the gap between need and solution should see significant uptake in on-line analysis[11] and it is in closing this gap that VISUM® will seize a very important market foothold.

MAIN ECONOMIC BENEFITS FOR THE USERS AND UNIQUE SELLING POINTS

The main economic benefits for users, and which would drive them to invest in the VISUM® solution, are derived from:

- VISUM will be compatible with the most standard production machines, devices and structures. SMEs will not need to change their existing equipment, devices or structures due to incompatibilities.
- Cost reductions due to reduced time, labour and cost intensive laboratory-based testing and manual quality inspection (which will also lead to higher throughput)
- High improvements in production performance; VISUM provides on-line information to the operators or automated machines to work continuously at the maximum efficiency
- Cost and space reductions by being able to remove/reduce stocks; companies will reduce (and in some cases completely remove) the need for lab analysis of raw materials and will be able to introduce the raw material in the production chain as soon as VISUM® verifies the pre-defined quality parameters are met.
- Savings through the effective elimination of end-of-line defects and faulty or contaminated products
- Savings through automated final inspection processes to increase production efficiency, preventing the costs associated with common or repeat mistakes caused by human error
- Economic returns through brand idealisation and increased market share by continually ensuring the high quality of the products and consequently protecting the brand.

If we consider the quantification of these financial savings through analytical headcount reduction, elimination of analysis time / instrumentation and reduced manufacturing cycle time, in the pharmaceutical industry as an example these could be greater than €400,000 for a typical pharmaceutical manufacturing facility with circa 30 batches manufactured per annum. Moreover, VISUM could go a long way towards reducing rejections in the pharmaceutical industry, the impact of which is in the order of billions of euro, especially as waste or rejection rate of 10%, unacceptable in most other industries, is considered good in the pharmaceutical industry (according to Centre for Drug Evaluation and Research or CDER)[12]. Batch rejects that result from process problems can be **higher than 20%**, the root cause being a **lack of process understanding and control**. Batch rejects can be a major issue as some APIs can be worth as much as $1million / kg.

[11] Robert N. Feudale, Nathaniel A. Woody, Huwei Tan, Anthony J. Myles, Steven D. Brown, Joan Ferre (2002). Transfer of multivariate calibration models: a review. Chemometrics and Intelligent Laboratory Systems 64 (2002) 181 – 192

[12] Contamination in the Pharmaceutical Industry. Risk News. Issue 18, February 2006

The Unique Selling Points (USPs) of the **VISUM platform vs. the current commercial offering**, that will spur users (SME decision makers, and Quality Manager and Production Managers from large corporations) to invest in a real-time monitoring solution are: ■ **Affordability** (~ €10-35k for a standard unit, which is highly cost competitive for bringing the power of a lab-NIR into the production line); ■ **Tailor-made solution** (whereby 'solution' means completely customised to the clients requirements and verified as 'fit for purpose'); ■ **Robust and reliable** (the offering of a tailor-made 'probe-sample' interface to overcome the limitations of generic systems that do not adequately fit/match each particular product line or monitoring challenge leading to mismatches that are the source of major errors); ■ **No biofouling** (via a novel layer of Thin PTFE specific polymers (non-polar compounds) coated on the surface of the optical window or an online sampler concept for difficult aqueous samples).

DESCRIPTION OF THE MARKET

Our VISUM IR-in/on-line monitoring solution is targeted at the European and **global food and pharmaceutical (initially), as well as chemical and other process industries**. These are high growth industries that are being influenced by global drivers such as the population increase, raw material demands coupled with the commercial need to deliver year on year growth and efficiency savings, and the increasing realisation that while emerging economies strive for growth through low cost manufacturing, western or established economies must compete through productivity and new technology[13]. In addition to competing, global economies must become more sustainable and less wasteful of natural resources in order to be sustainable[13].

Looking specifically at the **Food and Drink industry**, in the Europe alone, there are an estimated 310,000 companies generating an annual turnover in excess of €965 billion and providing employment to >4 million people.[14] Globally the industry is expanding rapidly to keep up with growing demand. Key to the competitiveness of the industry is the ability to maintain a high level of food safety and quality standards[15] in line with consumer, market and regulatory requirements. In order to improve the performance of food manufacturing processes and the quality of food products, there is growing industry interest in process sensors and in model-based monitoring and control methodologies[16]. This highlights the enormous market opportunity that is open for the adoption of the VISUM inline solution, especially as it will deliver an affordable, easy to use and reliable system to this highly price sensitive market. Improvements in process efficiency, reduced product variability, enhanced traceability, process understanding, and decreased risk of contamination are some of the benefits that will arise from the introduction of the VISUM in food processing. During the industry case studies that will be performed in 3 food processing companies through our Phase 2 Project, Return-on-Investment cases will be made.

Similarly, in the context of the **pharmaceutical industry**, which is one of the fastest growing industry sectors in the world today[17], EU and International[18] regulatory bodies are encouraging the drug lifecycle concept that links product and process development, qualification of the commercial manufacturing process, and maintenance of the process in a state of control during routine commercial production, whereby access to real-time on-line measurements is fundamental. The "Process Analytical Technology" (PAT) paradigm that is being promoted by the FDA (Food and Drug Administration in the US) and the EMA (European Medicines Authority) is based on improving final product quality by process design through knowledge of the fundamental scientific principles behind it, and continuous online control of a process. The **worldwide market is very lucrative for pharmaceutical producers and also for equipment suppliers, service providers and consultants**[19]. VISUM will play a major role as a PAT technology for pharmaceutical applications. The advent of generic drugs has changed the paradigm of high margins associated to IP rights; therefore, drug manufacturers have to base their profits on savings such as the ones enabled by PAT. Moreover, the drive towards continuous manufacturing in the pharma sector (which lags behind other industries, with significant emphasis on batch processing) lends itself to increased adoption of in-process monitoring and control systems, such as VISUM. The VISUM system could also have benefits for R&D in the pharma sector, whereby online analytics in pilot facilities could accelerate developments. Given that the cost of researching and developing a new chemical or biological entity was estimated at € 1,172 million in 2012[20], the drivers to invest in robust technology to reduce R&D costs are certainly there.

Finally, if we look at the chemical industry (which will be a post-project target market for the VISUM), in Europe the industry has a production value of €527 billion, within a worldwide total estimated by CEFIC as €3.1 trillion (2012). These industries contribute to European economy through an export share of 43%. Since the Process Analytical market supports this market growth, VISUM will have a knock on effect. While European sales values have doubled, its global market share has halved due to competition from developing low cost economies, hence there is a specific EU market need to become more competitive through technology. In conclusion, we are targeting lucrative and global markets through our initial (and post project) applications[21] and the drivers to invest in the VISUM open up an attractive business opportunity.

[13] Richard, J. Tweedie. Process Analytical Instrumentation, the challenges for in-situ characterization of Complex Particulate Materials. Procedia Engineering 102 (2015) 1714 – 1725. The 7th World Congress on Particle Technology (WCPT7) Process
[14] Nurturing the competitiveness of the European food industry (07.10.10) Enterprise & Industry magazine (http://ec.europa.eu/enterprise/e_i/index_en.htm), © EU, 2008 - 2010
[15] Commission Decision No 2008/359/EC of 28 April 2008.
[16] Process Analytical Technology for the Food Industry (2014). Editors: O'Donnell, Colm P., Fagan, Colette, Cullen, P.J. (Eds.). Springer (2014)
[17] http://www.efpia.eu/uploads/Figures_2014_Final.pdf
[18] FDA, Guidance for Industry, Process Validation: General Principles and Practices
[19] Pharma Plant Engineering: Trend To Disposables in Biopharmaceutical Manufacturing. Pharma Times-Vol 41- No.11- Nov. 2009
[20] Mestre-ferrandiz et al, Office of Health Economics, December 2012
[21] http://www.companiesandmarkets.com/r.ashx?id=55F9K2034700353

MAIN COMPETITORS AND THEIR COMPETITIVE SOLUTIONS

In the field of 'all-purpose' NIR lab-analysers, there are numerous big players such as Thermo, Foss, ABB, Sartorius, Bruker, PerkinElmer and Buchi, each with years of experience and a large customer base. Very recently, a number of manufacturers are offering NIR spectrometers sold as "out of the box" solutions (Table 7), which are marketed as solutions whereby the user only has to switch the equipment on, select an application, present the sample and press the run button and out comes the predictions. However, these solutions have been designed for solving common monitoring needs for wide markets and cannot be painlessly adapted to new more specific challenges, as they lack the necessary robustness in relation to the calibration models. Even in cases offering more flexible and non-specific configurations, the economical effort associated with the integration can be barriers for SMEs. For example, FOSS has recently launched the ProFoss in-line NIR system for dairy applications. However, it requires an off-line FOSS device to calibrate the in-pipe device. VISUM overcomes this through the automated/assisted, but high quality, **model builder**, which is a must in terms of providing affordable integration capabilities within non-specialised work.

TABLE 7- VISUM PLATFORM- COMPARATIVE ANALYSIS OF MAIN COMPETITORS

Manufacturer	Model	Target Applications	Built-in computing	Integrated chemometric tools	Built-in process control interface	Wireless comm.	Power autonomy	Antifouling solution	Price
Büchi	NIR-Online	All	No	No	No	No	No	Blower	50 k€
NDC	On-Line NIR	All	No	No	No	No	No	No	45 k€
Bühler	Multi Online Analyzer MYRG	Food-for Bühler machinery only	No	No	Yes	No	No	No	40-50 k€
Foss	ProFoss	Pharma and food (dairy)	No	Yes	Yes (OPC)	No	No	No	**55 k€**
Thermo	Antaris Target	Pharma (blending)	Yes	No	Yes	Yes	Yes	No	**45 k€**
Sentronic	SentroPAT FO	All	Yes	No	No	No	Yes	No	50 k€
Perten PerkinElmer	In-line DA 7300	Food (grains, flours)	Yes	No	No	No	No	No	**45 k€**
Bruker	Matrix-F	All	No	Yes	No	No	No	No	**50-80 k€**
Kemtrak	DCP007-NIR	All	Yes	No	Yes	No	No	No	30-40 k€
VISUM	**Online**	**All**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**10-35 k€**

From Table 7, we can see that VISUM is the most complete and flexible offering (the green cells highlight positive features) compared to competing solutions and aims to offer an excellent price-functionality/feature ratio: ■it is not closed in terms of applications; ■the instrument housing comprises the spectrometer and a standalone computer for generating the ultimate outputs to the production-line control system; ■a complete, updated and easy-to-use chemometric package is integrated into the built-in computer for allowing the users to develop their own models; ■ no prior background in chemometrics is needed as the user is guided step-by-step; ■ the computer can automatically generate actions –according to the predictions- in order to drive the process in a closed-loop; ■ the system can two-directionally communicate with external devices through a wireless channel; ■ the system is powered by batteries whenever needed – especially for standalone use on moving machinery; ■ the window is protected against fouling; ■ the system comprises a customizable online sampler if sample conditioning is needed (we will aim to develop a "template" suitable for some common cases to be defined in WP1. Such a template would be case-wise upgraded according to the client needs (in the commercialization stage).

MOST RELEVANT MARKET SEGMENTS FOR INTRODUCTION OF THE NEW SOLUTION

The VISUM will be fully deployed in our 'beta client' group (METALQUIMIA, DAIRYGOLDPASCUAL, SALGOT, ALMIRALL) during the Phase 2. These companies will serve as powerful endorsers of the VISUM® in the meat, dairy, olive oil and pharmaceuticals industry. Thanks to our ability to leveraging these application case studies (and to have lighthouse clients in each of these food segments and in the pharma industry), in Phase 3 Commercialisation phase (from Jan 2019 onwards) we will commence the active targeting of additional food segments and pharmaceutical applications in Spain, Ireland, and other selected markets as discussed below, and progressively from 2020 onwards we will enter other EU and global markets.

MARKET BARRIERS TO BE OVERCOME

The main barriers preventing uptake of NIR in the industry for in-process monitoring are complexity and cost of integration, and calibration and model maintenance. Evidence-based application notes (including ROI quantifications), whitepapers and publications from the project will be powerful documents for demonstrating how VISUM® overcomes these technical barriers. From a regulatory context, the most significant market barriers to entry will be standards and validation requirements and in particular in pharma applications. In Phase 1 we analysed all regulatory requirements with our first beta clients and as such we will ensure pre-validation according to regulations (e.g. CFR 21, Part 11…) as discussed in Section 2.2).

TARGET USERS AND HOW THEY WILL BE REACHED

The targeted users of the final VISUM solution are food, pharmaceutical (and chemical companies in the post-project phase), including from SMEs to large enterprises. This is a global and growing market. Our first effort will focus on actively targeting the **headquarters**

(HQs) of large-size companies in the Food, Pharma and Chemical industry, which have been defined as **the early innovator group**, due to a) their high production volume justifying an investment in an in-line monitoring solution; b) the probability of follow-on contracts is high for large-size companies: once first in-line monitoring units are successfully tested, the probability is high that they ask for further units to roll-out inline monitoring in their international production plants. VISUM will establish direct contact with HQs as they are usually in charge of implementing new technologies and process improvements. The HQ locations of the Top100 companies in the food[22], pharma[23], and chemical[24] (3) industry are presented in Table 8. The majority of HQs for all three sectors are located in the EU (mainly Germany, Switzerland, France, UK and Italy), North America (US and Canada) and Asia (mainly Japan and India, and China).

TABLE 8- HQ LOCATIONS OF TOP100 FROM FOOD, PHARMA AND CHEMICAL PROCESSING INDUSTRIES

	FOOD		PHARMA		CHEMICAL		FOOD + PHARMA + CHEMICAL	
Area	**% Area**	**Main countries**	**% Area**	**Main countries**	**% Area**	**Main countries**	**% Area**	**Main countries**
EU	31%	FR(7), DE(5), UK(5)	36%	DE(6),CH(6), FR(4), IE(4), IT(4)	30%	DE(11),CH(7)	32%	DE(7),CH(5), FR(5), UK(3), IT(3)
North America	41%	US(37), CA(4)	23%	US(22)	32%	US(29)	32%	US(29), CA(3)
Asia	21%	JP(16), China(3)	38%	JP(26), India(7)	34%	JP(15), SKorea(5)	31%	JP(19), India(3)
South/CentralAmerica	6%	BR(3), MX(3)	1%		3%		3%	
Other	1%		2%		1%		2%	
TOTAL	**100%**		**100%**		**100%**		**100%**	

Selling VISUM INLINE involves selling the hardware equipment, the device calibration (acquiring spectra of client samples, conducting laboratory referential analysis and elaboration of predictive models) and integration in the production line (engineering). Due to a relatively high service proportion, direct contact with important clients will be crucial. For this purpose, IRIS has plans to open regional offices (and we already have offices in Dublin, Barcelona, New York and San Francisco) to acquire high-priority clients, establish agreements with distributors, manage distributors and deliver the service part during and after the acquisition process. Distribution roll-out will take place in two phases as is described below under the heading *'Strategy plan for commercialisation of the business innovation project'*. IRIS has an experienced Sales and Marketing Department and the experience to effectively reach the decision makers (Plant Managers, Quality Managers, Production Managers, R&D and Innovation Directors) in the process industries is present. Targeted communication campaigns to companies from VISUM existing client-base and networks (which include such large industries as GSK, SANOFI, ALMIRALL, DSM, P&G, MEGARA RESINS, DAIRYGOLD, NESTLE, DAWN MEATS, among others) and new prospective customers will be designed (via electronic mailings, paper mailings, personal/face-to-face communication…), and through the channels of our distribution/sales agents (discussed below). Prospective customers will be found using a combination of technical presentations at conferences and exhibitions and our website with application note/examples, among others (listed in the Communication section below).

B) COMPANY

RELEVANCE, RATIONAL AND ALIGNMENT OF THE INNOVATION BUSINESS PROJECT WITH OUR BUSINESS STRATEGY

The VISUM innovation project is at the core of our strategy to becoming a world leading provider of advanced monitoring solutions for the food, pharma and other process industries. Since its beginnings, VISUM has been active in the transitioning of spectroscopic lab analysers to on-line applications for real-time monitoring within the framework of European R&D projects, and as part of B2B bespoke solutions for food and pharmaceutical companies in Spain and Ireland. Based on extensive engagement with industry over the years, we have identified a clear market opportunity for **customised monitoring technology that shifts the emphasis from "device" to "fully integrated solution"** based on tailoring the device to the line and to the specific operating environment of the client. Central to the ability to mainstream customised and affordable monitoring solutions for industry, is a flexible and reliable technology platform based on state-of-the-art low-cost MEMS spectroscopic probes, combined with built-in data management software and the latest in chemometric model advancements for facilitating rapid calibration of the VISUM hardware for easy set-up and customisation to the client monitoring needs and industrial environment. The VISUM solutions are a major opportunity for us to grow our company, which specialises in spectroscopy solutions. The project has the full backing of our Management Board and is considered as a strategic project for advancing our technology development roadmap, the implementation of which is intrinsic to ensuring the future growth of our company. The Board has approved further internal funds for VISUM ® for investing the necessary 30% co-funding required to supplement the SME Phase 2 Instrument and ensuring we have funds to cover the commercialisation phase. A Business Model Canvas has been provided in Annex 2 (which summarises our current business plan, which will be built on through Task 5.2).

GROWTH POTENTIAL OF THE SOLUTION

During the Phase 1, we dedicated considerable effort to developing our business plan for the VISUM® product. During this period, we engaged in extensive field research, whereby we attended exhibitions and conferences (BTA Barcelona Food Technologies, Global Food Safety 2015, ACHEMA 2015, Anuga Food Congress, PharmaProcess, IFPAC), made individual company visits and conducted face-to-face meetings with future clients. We also attended the Photonics West Exhibition in San Francisco earlier in the year in order to review

[22] http://www.foodengineeringmag.com/global-top-100-food-&-beverage-companies
[23] https://www.google.com/fusiontables/DataSource?docid=1uEZuNGJlP68Imym7zAVT-XHXwhHdvs2lT5L6iNs#rows:id=1
[24] ICIS Chemical Business

current spectroscopy technology and others, such as fluorescence. We also brought a world renowned NIR expert, Dr. Emil Ciurczak[25] (widely known as the 'NIR Prof' and internationally recognized) over to us from New York in June 2015 in order to spend 3 days mentoring senior management in relation to market strategy and the business aspects of NIR-based solutions commercialisation, whereby he advised us on such aspects as pricing, and gave expert insights into the status of Process Analytical Technology adoption in the industry, as well as current regulatory frameworks. He also provided training courses to our engineering staff in aspects such as engineering challenges, sampling, integration, etc. All these experiences and the feedback that we have had from industry have strengthened our belief in the VISUM offering and the associated business opportunity.

Since 2011 we have commenced our strong specialization in spectroscopy-based advanced monitoring solutions by developing ad hoc solutions for clients in the food and now more recently the pharmaceutical industry. The VISUM platform concept is born out of our industry experience and an understanding of the challenges and opportunities in relation to a revolution in real-time monitoring and advanced control that is undoubtedly going to take place as Industry 4.0 concepts take a real foothold in the process industries- which primarily include food & beverages, pharmaceuticals, chemicals, oil & gas, among others. Increasing market demand for high quality, consistent or uniform end products in process industries, as well as for value added and customized products (which involve increasingly complex processes and also require increasing flexibility in manufacturing procedures), as well as industry need to reduce downtime, and to ensure safety and security for plant operators, are all fuelling the growth of the Sensing, Monitoring and Control market. The Global Sensing, Monitoring and Control market was estimated to be worth €377 BN in 2015, with the European market being worth €122 BN[26]. With a projected growth of 7% per year in the Monitoring and Control segment and 6% in the Sensing segment, projections to 2020 estimate the Global Market to reach €524 BN by 2020, with Europe reaching €171 BN. These volumes are highly achievable, in view of a huge and growing market opportunity for robust, reliable, affordable, and readily integratable IR spectroscopy based chemical monitoring solutions.

We have defined our sales target for the first 5-year commercialisation period (i.e. from 2019 to 2023). Over that period, we aim to install ~1000 units in processing lines/facilities in food, pharma and other process industries in Europe and globally (Table 8). Based on market analysis work we performed during the Phase 1 project, we can confirm that this is an achievable and realistic target, considering the strong industry demand for real-time monitoring and QC systems (and also considering that some of the larger clients will acquire multiple units for multiple lines, processes and sites. It is expected that the global market will offer high growth potential for the medium to long-term as our international distribution networks kick in (with high growth markets in the longer term being Asia where increasing processing activity and rising standards will drive market demand for affordable solutions), whereby we expect a situation beyond 2023 that will see 2:1 international vs. European sales. The maintenance contracts and chemometric upgrades (known as 'models maintenance') will also be an area for sustained growth, as will revenue secured through offering data analytics services (to mine the historical data stored on the cloud).

By 2023, we expect to have generated a cumulative total (Table 9) of €29.25M from unit sales and €13.35 M in service contracts (data services, model development, maintenance contracts, etc.). These figures have been calculated based on an average sale value of standard installed VISUM solution being €22.5k. Revenue will also be generated from Lifetime Maintenance Contract costs per unit (based on typical averages for industrial analytical instrumentation), and income from data services (data analytics contracts and model building), which will be a high growth area for the future. Based on a present day turnover of € 4.8 M in IRIS, VISUM® will allow IRIS to increase its turnover >4-fold by 2023 and grow its workforce by an additional ~39 staff. An average 35% gross margin will yield gross profits for period of almost €13.7 M, which will provide revenue for further investment in technological improvements and new applications for further growth in relation to widening the range of specifications for accessing to a larger variety of applications. In this sense, it is envisaged that the VISUM specs could be extended in terms of higher and lower environment temperature limits (e.g. from -20 to 90 C), sampling rate (faster, e.g. <1 mS acquisition time), and simultaneously keeping or improving S/N- to be used in intensified processors, which require faster PAT tools, wider wavelength range (from SWIR (0.0-1.7 um) to E-SWIR (0.9-2.5 um)), ability to acquire more chemical information and consequently extending the applicability to more "difficult" samples and, finally, more sophisticated chemometric tools, by including the latest mathematical achievements derived from the development of machine learning (artificial intelligence) state-of-the-art, with algorithms such as SVM (support vector machine) and SGD (Stochastic Gradient Descent).

TABLE 9- VISUM UNITS DEPLOYED IN EU AND GLOBALLY & REVENUE GENERATION 2019-2023

	2019	2020	2021	2022	2023	Total	Jobs	Margen
Number of units sold in Europe	20	60	130	230	360	**800**		
Number of units sold in US	5	25	70	140	260	**500**		
@€22.5k)	€ 562,500	€ 1,912,500	€ 4,500,000	€ 8,325,000	€ 13,950,000	**€ 29,250,000**	27	€9,371,250
Service contracts: data servcies, additional models, software upgrades, maintenance contracts	€ 168,750	€ 669,375	€ 1,800,000	€ 3,746,250	€ 6,975,000	**€ 13,359,375**	12	€4,382,438
Total	**€ 731,250**	**€ 2,581,875**	**€ 6,300,000**	**€ 12,071,250**	**€ 20,925,000**	**€ 42,609,375**	**39**	**€13,753,688**

[25] Emil Ciurczak since works with IRIS as an in-house consultant and his CV is included in Section 4.1

[26] Jorge Perieia (2007). Monitoring and Control. Today's market and its evolution till 2020. Final Report of the Study SMART 2007/047. European Commission Information Society and Media.

HOW THE COACHING SERVICES OF THE SME INSTRUMENT WILL BE USED

During the Phase 1 project, experts at Enterprise Ireland (EI) performed an Innovation Diagnostic on our company that yielded above average results (which was to be expected based on the vast innovation management planning and continuous improvement approach of our company in the preparation for securing- and now for maintaining - **ISO 9001:2008 and UNE-EN-ISO 166002: Management Systems R&D&I certification**). The VISUM Management team in conjunction with the EI experts, consider that coaching services focussed on commercialisation activities would be very valuable. This coaching would focus on defining the 'Compelling Value Proposition' and the Promotion, Placing and Pricing aspects. This information would help design and drive a focussed marketing campaign for VISUM®.

FUNDING REQUIREMENTS FOR REACHING COMMERCIALISATION STAGE

There are 3 key stages to the commercialization: 1) **Start selling** (we will leverage the Phase 2 project to gear up to this so that first sales can commence in the immediate post-project phase); 2) **Expand the sales effort and applications support and rapidly increase distribution network**; 3) **Going global**- creating a network of centres and support. Table 10 provides an overview of the costs in the **1st year post-project**. The financial mix will be own funding and mainly revenue from first sales, whereby based on targeted sales of 25 units (i.e. 2019), we will generate a sales revenue of an estimated €731k. However, if first year sales to do match targets, our financial mix will be own funds combined with other external funds (bank loan, regional funds (e.g. ESIF smart specialisation funding) or an investment round if needed).

TABLE 10- FUNDING REQUIREMENTS FOR REACHING COMMERCALISTAION PHASE

Setting up the manufacturing and warehousing facility (conditioning work in IRIS facilities)	€35,000
35 units will be built in the post-project year to serve first orders and build up a stock inventory (average manufacturing cost is 6k/unit)	€210,000
Additional costs associated with securing certifications, validations, etc. (ISO, ATEX, CE mark)	€15,000
Building the extended sales and distribution channels	€22,000
Product marketing/promotion (including product launch and exhibitions)	€80,000
Investment in the first sales effort (travel client visits, sales support costs, e.g. lab tests, etc.)	€30,000
Staff costs (full and part-time production team, application team, sales team, management team)	€500,000
Overhead costs (contribution to rent, admin, utilities, etc.).	€100,000
Shipping, packaging, insurance costs for deliveries to serve orders	€6,250
Expected installation costs associated with first year units sold (approx.25 units envisaged in 2019)	€58,500
After sales support on units installed	€29,250
Total costs in first year of commercialisation	**€1,086,000**

2.2 MEASURES TO MAXIMISE IMPACT

A) DISSEMINATION AND EXPLOITATION OF RESULTS

KEY STAKEHOLDER INVOLVEMENT FOR SUCCESSFUL EXPLOITATION

The following stakeholders have been identified for their possibility to add value and to facilitate the commercial exploitation of the VISUM® platform: ■ **Additional beta clients** from different food segments, pharmaceutical processing (and chemical industries) who will enable us to create additional application studies and Business User Cases; ■ **Reputable and effective distributors and sales agents** for building our European and global commercial network, along with **OEMs** (e.g. Metalquimia) that could be interested in integrating VISUM into their equipment to create a 'smart' or 'intelligent' equipment offering; ■ **Industry Associations and Clusters** who can provide a very powerful pathway to their members. They also enable active engagement with policy makers and in helping the industry and policy-makers to work together towards common goals in relation to process efficiency, quality, safety, sustainability, etc. ■ **MEMS-NIR sensor providers**, whereby VISUM® provides them with an integration partner to mainstream the adoption of their sensors in in-process industry applications. ■ **Equipment certification consultants** to facilitate all the steps involved in obtaining certifications; ■ **Renowned academics and industry influencers** who can independently test the VISUM platform and publish/disseminate the results.

FURTHER STEPS NEEDED BEFORE THE RESULTS ARE MARKET READY

Our approach to the planning the steps to being market ready has centred on the well-established **4 Ps Marketing Mix- Product, Price, Promotion, Place** (Table 11), which we have used as a framework to map out the necessary actions and measures to ensure we are well prepared to successfully reach the market with our VISUM product (based on past experience we know this is a robust and complete approach to planning a pathway from product to market). Through *WP5-Gear up to market*, we will implement, build on and further detail plan of activities.

TABLE 11- 4 PS MARKETING MIX APPLIED TO THE VISUM PHASE 2 AND THE POST-PROJECT PHASE

4 Ps Marketing Mix	Related Project activities	Related Post-project activities
Product (product functionality and features, services, brand, etc.)	■ Structure the value proposition of our new product (to validate and improve our value propositions we have formulated). ■ Draw up the product datasheets that will outline the key information	■ Plan and implement a Roadmap of product/service improvements and additional features for effective product lifecycle management

	■ Create product brochures outlining applications (the applications that have been validated in this project and additional applications), highlighting the benefits, etc. ■ Define the services that support the product (maintenance service, customisation services, upgrades to the software and models, data services and consultancy based on sensor use and data understanding). ■ Brand building activities to strengthen brand identify.	■ Implement new applications development plan (additional foods and beverages, additional pharmaceutical applications, chemical industry, bioprocessing, etc.)
Price (price lists, discounts, margin, credit terms, etc.)	■ Finalise and validate 2019 price list based on our manufacturing and logistics costs, required margin to have a profitable business, necessary margin for distributors/sales agents, and desired pricing positioning vis-à-vis market tolerance and competitors (we have initially estimated the expected price for an average installed unit at €10-35k- based on our existing technology and market experience). ■ Define our pricing policy in relation to discounts, credit terms, etc.	■ Review and maintain pricing.
Promotion (Communication, Advertising, Sales Promotion, etc.)	■ Exhibition at 4 tradeshows (e.g. Alimentaria, FoodTech, ANUGA, ACHEMA, EXPOQUIMIA, etc.) ■ Presentations at 2 exhibitions/conferences (e.g. Global Food Safety Conference, Annual European Food Manufacturing & Safety Summit, Pharma Process, APACT, EuPAT…); ■ Publishing at least 4-6 articles in industrial/sectoral channels (technical/sectoral magazines, etc.) and in peer reviewed journals (2-3). ■ Publishing at least 2 whitepapers and at least 4 application notes on the product page of our website, etc.; ■ Write 4 own blogs that will be published on our website and promoted via our social media accounts and write at least 2 feature blogs as guest bloggers on relevant on-line sources ■ Design and implementation of a social media campaign (whereby we will for example publish regular tweets- at least 1 per week, whereby we are disseminating our own content or related content of third parties- e.g. content in relation to product quality and safety, Industry 4.0, PAT, etc.). ■ Preparation of at least 3 press releases over the course of the project (one at project launch and 2 more to coincide with key milestones). ■ Creation of promotional materials (webpage, leaflet, posters, banners, QR code, video and virals, etc.)	■ Organise official product launch (along with a press conference) ■ Roadshow of exhibitions and tradeshow ■ Presentations at conferences ■ Publishing regular articles in industrial/sectoral channels and in peer reviewed Journals ■ Publishing regular whitepapers and additional application notes on website, etc.; ■ Advertising via digital ad campaigns ■ Sponsorship of selected industry focussed conferences/events; ■ Active PR campaigns (e.g. whereby we will aim to reach out to industry influencers who could be a powerful source of reviews) ■ Regular publishing of feature blogs as guest bloggers in on-line sources; ■ Sales promotion: e.g. we will use this tool for our launch campaign- discount for companies that place orders during the launch period, sales promotions linked to exhibitions, etc.
Place (Distribution Channels, Coverage, Inventory, Logistics, etc.)	■ Establish manufacturing and associated logistics (Task 4.4)-the first 5 product units will be produced and quality assured. ■ Establish and train internal Sales Team (Our sales staff will play a major role in promotion (see above) of the product, and in the Phase 2 they will be trained fully on the VISUM. ■ Establish Application Team (who will support the Sales Team) ■ Set up First Tier Distribution network (identification and negotiation) ■ Establish agreements with shipping companies for transportation and insurance companies	■ Build product stocks ■ Set up warehousing inventory ■ Set up second tier distribution network

STRATEGY PLAN FOR COMMERCIALISATION OF THE BUSINESS INNOVATION PROJECT

Our **commercial strategy** is to commercialise the VISUM system through **own commercialisation**. Marketing process sensors is a specialist business. In-situ process sensors typically have an extended sales pipeline, with considerable upfront customer interaction, evaluation and testing, and due to the nature of the process manufacturing industries, process analytical systems require some level of customisation, and they also need to be integrated (and especially as we are offering a turnkey solution) and ongoing support has to be ensured. Even while the VISUM platform will facilitate the integration work (and this is one of its Key Value Propositions), some degree of installation and ongoing support has to be ensured. As such, there are two key aspects of the business plan: a) the creation a **specialist Application Team** (to support Sales) and, b) the upfront development of a **Network of Distributors and Agents** to open up sales in European and international markets.

Under the leadership of the CEO, the core **Sales Team** in VISUM will be nominated and trained. In parallel, the **Application Team** will be created and will be composed of dedicated engineers who will provide support to the sales team by testing materials (e.g. samples from prospective clients and for building new application notes), performing detailed site visits, where required, providing advice and support during system installation, providing training, and handling both site and remote customer support. In relation to our own current and planned geographical presence (figure 3), VISUM currently has production and commercial facilities in Ireland (which allows us to serve UK as well) and Spain (which enables us to serve Portugal and Italy), and as part of our further internationalisation strategy we

have recently opened technical/sales offices in US East Coast (New York City) and West Coast (San Francisco). Our plans for 2016/2017 are to open offices in Germany (that will also serve Switzerland and Austria), France (which will serve Benelux), Japan (for our Asian presence) and Mexico. In relation to establishing a network of **distributors and agents to help us build sales, and to provide a local/regional point of contact for customers and follow up on the sales process**, we will focus first of all on securing distributorships (and agent agreements) in the regions where we have identified as Priority 1 by end of 2018 (1st Phase), i.e.: Europe, US, Japan and Mexico. This work will involve identifying and negotiating agreements (e.g. exclusivity or not, margins, etc.). By 2020, we will expand to include Distributors in other key markets. Our distributors and agents will initially be supported by our own internal Sales Team and Applications team. However, the longer-term plan is to see these relationships being long term partnerships and ultimately growing these into regional centres of expertise able to deliver complete solutions. Hence over time these regional centres will develop local application teams and the local sales support will be complimented by the establishment of **regional integration agreements with engineering companies** to directly support our commercial and technical centres. Agreements will also be established with global OEMs for delivering built-in VISUM systems in their equipment (e.g. METALQUIMIA, GLATT, GEA among others).

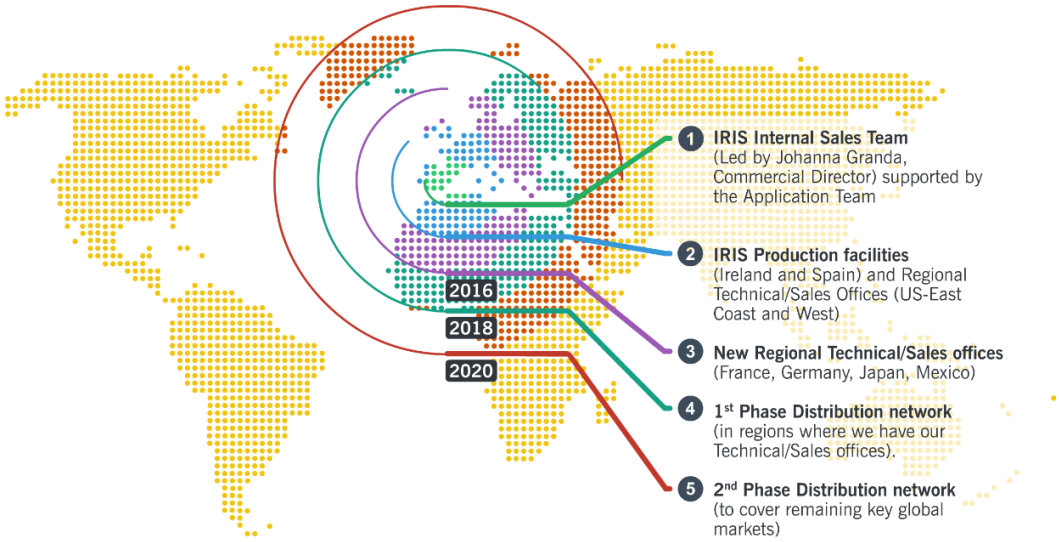

FIGURE 3- TIMELINE FOR IMPLEMENTATION OF COMMERCIAL STRATEGY

Time to first sales is Jan 2019, whereby we will commence the sale of first units to our beta clients in the immediate post-project phase and we will build out our sales effort from there. The end-user involvement in the Phase 1 and Phase 2 of this Innovation Project, through our 'beta clients' or early adopters from the food and pharmaceutical industry will provide a quick route to first sales if industry trials at each site are successful.

B) INTELLECTUAL PROPERTY, KNOWLEDGE PROTECTION AND REGULATORY ISSUES

INDUSTRIAL PROPERTY RIGHTS ASSETS

VISUM owns the VISUM® brand and all the know-how behind the development to date. VISUM will **own all the knowledge items that will be generated through this SME Instrument**. We have agreed to grant our early adopters (METALQUIMIA, SALGOT, DAIRYGOLD, SALGOT) with preferential use of the results (free industrialisation of the first VISUM system installed in their plants in exchange for them acting as validators of the system). In relation to our freedom to operate, we have performed periodic patent reviews in parallel with our developments to date, which have all concluded that no existing patent directly competes with the envisaged VISUM® system as a whole.

MEASURES FOR ENSURING COMMERCIAL EXPLOITATION

As a measure for ensuring commercial exploitation of the project results, a systematic Technology Watch task will be implemented over the course of the work plan (through T5.1) to ensure we keep up to date of relevant competing developments. In addition, every 6 months a patent review will be performed to reconfirm our 'freedom to operate' and to ensure we are not infringing on the IPR of a third party. A SWOT analysis will be prepared at M6 of the project by the VISUM Management Board and it will be reviewed and updated each quarter in relation to the VISUM ® solution strengths and weaknesses and in the context of the market opportunity and any threats to realising commercial exploitation. A major effort we have made in order to pave the way forward for commercial exploitation has been lining up a powerful group of "early adopters", through our 4 first beta client companies of different sizes, representing different segments of the food industry (dairy, olive oil and meat) and the pharmaceutical industry. These will allow us to gain serious endorsement value from the Phase 2 project and to document impactful case studies/business cases in the use of VISUM ® for in-process monitoring applications.

KNOWLEDGE MANAGEMENT AND PROTECTION STRATEGY

A patent search has been performed to confirm our freedom to operate. The last patent search was performed on March 21st 2016 using the World Intellectual Property Organisation (WIPO) PATENTSCOPE database (https://patentscope.wipo.int/search/en/search.jsf) using the key words: ALLTEXT: NIR + measuring; ALTEXT: process analytical technology; ALLTXT: online probe spectroscopy. The results revealed that no patent exists that would affect the VISUM development. In fact, many of them are associated to specific processors

such as drum coaters, sorters as well as specific bodies -such as tablets- or substances -such as flour or crude oil- while VISUM is intended for generic applications no matter the specific nature of the processor or the product to be monitored. Likewise, a relevant number of patents are associated to Raman spectroscopy while VISUM is designed on the basis of IR spectroscopy. In this set of patents, the following are particularly relevant: FOSS ELECTRIC A/S: EP0935749: A METHOD AND FLOW SYSTEM FOR SPECTROMETRY AND A CUVETTE FOR THE FLOW SYSTEM, which is particularly focused in milk and dairy products; BUEHLER AG and PERTEN PETER: EP0539537: PROCESS AND DEVICE FOR THE IN-LINE NIR MEASUREMENT OF POURABLE FOODSTUFFS, which is focused in grains. However, there are some patents that are too generic because many of their claims were not novel enough at the application date. A further in-depth analysis of possible conflicts should be undertaken. As examples of such kinds of patents: WO1999058959: NEAR INFRARED SPECTROMETRY FOR REAL TIME ANALYSIS OF SUBSTANCES; WO2000020834: MULTI-COMPONENT PROCESS ANALYSIS AND CONTROL. Basically, we can conclude that there are no evident IP conflicts between VISUM and existing (patented) similar systems because, on the one hand, the most of the related patents are constrained to specific applications and, on the other hand, IR spectroscopy has been used during the last 50 years for process monitoring; there is no room for novelties in terms of IP blocking.

In relation to the novelty of our VISUM development, in line with the design and development work that will be performed in the Stage 2 we will analyse all outputs for their novelty and for their protectability through Patents, Trademark, Industrial design, Copyright, Trade-secret, or Confidentiality, as appropriate (i.e. the form which offers the most adequate and effective protection for each type of result). We envisage that we will gain IPRs through a number of aspects of the development (all of which will be confirmed and pursued as the developments unfold): ■ The designs for the self-calibration feature, whereby the specific combination of optomechanical solutions with ad hoc procedures could be protected as key knowhow; ■ The designs for a universal sampling unit will imply developing novel setups aimed at facilitating plug & play features which are not currently available; ■ The optical setup will be specifically designed for optimizing signal/noise ratio and reducing the drawbacks associated to scattering due to the sample nature; such a novel design will be also protected in some way (knowhow or patent); ■ The self-calibrating models, which will be protected by copyright in Europe and could be patent protectable in the US will necessarily involve radically new approaches in terms of artificial intelligence (deep learning) tools that have not previously used in traditional chemometrics. As addressing the ideation of such solutions will be tackled according to the TRIZ methodology, it is envisaged that some of the outputs could be novel and specific enough for being patented. In fact, one of the advantages of using such a methodology is its productivity in terms of novelties. During the Phase 1, we submitted a trademark application for protecting the VISUM brand name and its logo. During the Phase 2, we will study deeper the registration of the MINI-PAT concept, which is consists in having a complete set of basic PAT (Process Analytical Technology) tools integrated into the one instrument whereby the probe is designed for plug and play use. The MINI-PAT concept has the potential to be the Holy Grail of the PAT movement in industry. It is our intention to also register the “Get in line” commercial tag line.

REGULATORY AND/OR STANDARD REQUIREMENTS TO BE FULFILLED

All EC and national regulations that could influence on the commercialisation of the project outputs will be studied and their influence on the development or the future use of VISUM platform will be taken into account. The requirements in relation to adherence to: ASTM E1655-97; ASTM E1866-97; ASTM E1790-04; IUPAC Guidelines for Calibration in Analytical Chemistry; EMEA/CHMP/CVMP/QWP/17760/2009 Rev2; 11 CFR Part 35; 21 CFR Part 11, EC Machinery Directive and food-processing equipment (EU 98/37/EC Directive), etc.) and standards (e.g. EN45011 or ISO65 quality standards and all manufacturing standards set by ISO and to the GMP regulations, as well as to compliance with the CE Mark, will be given particular attention during the design stages of the system. ATEX certification requirements will also be fulfilled. Through Task 1.2, the regulatory and standard requirements will be studied with the end-users to ensure that all necessary actions that need to be performed during the development work in Phase 2 are carefully planned. During Task 4.1, certification applications will be made and the appropriate steps will be initiated towards obtaining approvals by the standards bodies and to ensure compliance with guidelines, standards and regulations so that VISUM® can be sold and used in food and pharmaceutical processing in Europe and internationally.

MEASURES TO PROVIDE OPEN ACCESS TO PEER-REVIEWED SCIENTIFIC PUBLICATIONS

It is our intention to publish aspects of the project results in peer-reviewed scientific publications, and these will be made available free of charge through Green Open access publishing. Our intention is to have publishable results from the validation of the chemometric models from T1.4, among other publication opportunities.

C) COMMUNICATION

In line with the Promotion part of our 4Ps Marketing Mix described in Table 11, the communication activities that will be performed over the course of the project will have the principal objective of **communicating the benefits and value of our VISUM product to our future clients**. As such, a big emphasis will be on communication activities to increase awareness about the product and its benefits to our target segments. We know from past experience that the strongest promotional tool is **successful applications**. Early testing provides material for early promotion. Building our product development around ‘beta clients’ gives the product the evidence to prove

capability and the push to have strong growth from day one. This will provide us with engaging and powerful content for our communication campaign towards our future clients.

We intend to build the brand identity and promote the product actively through standard means such as the project web, leaflet, poster, etc., as well as through more vanguard and dynamic tools such as blogs, social media (LinkedIn and twitter), Infographics, etc. We will also produce white papers as an impactful communication tool. Digital marketing will also be employed for reaching users. In Internet marketing, storytelling is a potent method of converting sales prospects to customers. As such, we will use the 'business stories' technique, whereby we will document case studies from our early adopters (no sensitive processing information will be disclosed and only relevant non-confidential information will be used) to communicate the business benefits of investing in the VISUM system in an impactful, creative and engaging way in order to effectively connect with industry decision makers. We will showcase and exhibit results at 4 exhibitions/tradeshows over the course of the project (e.g. Alimentaria (Barcelona), FoodTech- formerly BTA- (Barcelona), ANUGA (Cologne), ACHEMA, EXPOQUIMIA, ANALYTICA etc.) and in at least 4 conferences (e.g. Global Food Safety Conference, Annual European Food Manufacturing & Safety Summit, Pharma Process (Barcelona), APACT, EuPAT, etc.). We will also publish 4-6 articles in industrial/sectoral channels (technical/sectoral magazines, etc.). Publications of non-confidential and protect results in peer reviewed journals (e.g. J. of Spectroscopy, J. of Food Processing, J. of Process Control, J. of Process Analytical Chemistry, etc.) are considered as very powerful marketing tools to the industries, such as pharma, food, chemical and other process industries, as well as to policy makers and standards bodies, and will aim for at least 2-3 papers resulting from the project. We will mobilise our membership in networks and cluster memberships to promote the VISUM solution and to participate in events/workshop, etc. that would enable us to present VISUM and disseminate the successful case studies: SPIE (International Society for Optics and Photonics), A. SPIRE (Sustainable Process Industries), SECPHO (Southern European Photonics Cluster), among others. In short, our communication activities will ensure we maximise the future dissemination and exploitation of the project results.

Finally, as we are benefitting from tax-payers contributions, and given the high societal relevance of KETs that help to ensure food/pharma quality and safety, there is scope through social media to make the general public aware of the VISUM development and the associated societal benefits, e.g. contribution to improved industrial competitiveness, job creation, prosperous regions, access to high quality, safe and affordable products, the ability to enjoy healthy environments; these are all benefits that contribute to our quality of life as EU citizens.

3. IMPLEMENTATION

3.1 WORK PLAN – WORK PACKAGES, DELIVERABLES AND MILESTONES

OVERALL STRUCTURE OF THE WORK PLAN

We envisage a 24-month work plan and 6 work packages as follows (The Gantt Chart (below) and the Pert (Figure 4) show the timings and interdependencies between the different WPs and tasks): **WP1** will commence by preparing samples from each of the end-users so the feasibility prototype system available in VISUM we will undertake an initial demonstration of the system on these applications to provide test data for the model development. An integration strategy for each installation will also be defined. Based on the system specifications that were defined in a co-creation approach with these end-users (and output from the Phase 1 project), designs of the VISUM® system hardware and software layers can commence **(WP2)**, and the hardware prototypes units will be manufactured. The integrated VISUM hardware and software will be integrated and pre-validation tests will be performed, prior to integration in the industry. The VISUM prototypes will then be installed and beta tested on the selected applications at the end user sites by the end users and supported by the development team to validate the design **(WP3)**. Having validated the design, we will translate the prototype to a product **(WP4)** through the creation of the manufacturing documentation package using the engineering information and specifications. This will be passed to purchasing and production engineering to enable manufacture. Deployment of these first products will be in our participating end-users and to additional selected Beta customers to enable early refinement and

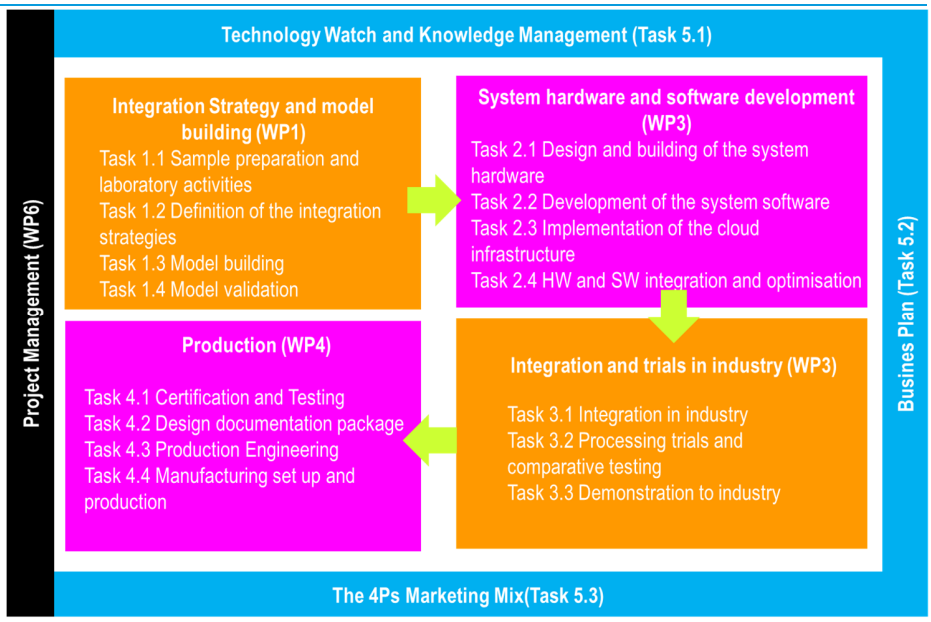

FIGURE 4- PERT DIAGRAM OF WORK PLAN

troubleshooting support by the Application Team. In parallel with the product development work, we will further develop our business development plan **(WP5)**. As part of this activity the 4 Ps marketing plan will be implemented and we will build our internal Sales Team, Application Team, along with the first tier of our international distribution network for reaching key markets. Finally, a project management framework will be implemented through **WP6** and the 1-page Gantt is an effective tool for high-level project control and a helicopter overview for effective project management.

TIMING OF THE DIFFERENT WORK PACKAGES AND THEIR COMPONENTS

No	WPs and Tasks	Del. & Milestones	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	Integration strategies and model building	ML1, ML3			#						#																		
1.1	Sample preparation and laboratory activities																												
1.2	Definition of the integration strategies	D1.1			■																								
1.3	Model building	D1.2, D1.3							■		■																		
1.4	Model validation	D1.4									■																		
2	System hardware and software development	ML2, ML4, ML5						#				#		#															
2.1	Design and building of the system hardware	D2.1, D2.2				■		■																					
2.2	Development of the system software	D2.3										■																	
2.3	HW and SW integration and optimisation	D2.4, D2.5												■■															
3	Integration and trials in industry	ML6, ML7														#				#									
3.1	Integration in industy	D3.1														■													
3.2	Processing trials and comparative testing	D3.2, D3.3																		■						■			
3.3	Demonstration to industry	D3.4																											■
4	Production	ML8, ML10																			#							#	
4.1	Certification and Testing	D4.4																											■
4.2	Design documentation package	D4.1																			■								
4.3	Production Engineering	D4.2																			■								
4.4	Production	D4.3																										■	
5	Gear up to commercialisation	ML9																										#	
5.1	Technology Watch and Knowledge Management	D5.2																		■									
5.2	Business plan	D5.3																									■		
5.3	The 4Ps Marketing Mix	D5.1, D5.4												■													■		
6	Project Management																												
6.1	Monitoring and reporting	D6.1, D6.2																											
6.2	Quality and Risk Management																												
6.3	Innovation Management		■		●			●			●			●			●			●			●			●			
6.4	Fulfilment of Contractual Requirements		◊		■			■			■			■			■			■			■			■			

■ Deliverable / # Milestone / ◊ Project Kick Off Meeting● Meetings of the Visum Ltd. Management Board

GRAPHICAL PRESENTATION OF THE COMPONENTS SHOWING HOW THEY INTER-RELATE

See Pert Diagram (Figure 4) in the work plan introduction.

3.2 MANAGEMENT STRUCTURE AND PROCEDURES

■ ORGANISATIONAL STRUCTURE AND DECISION-MAKING- As this is a single entity project, the organisational structure (Figure 5) is practical and simple, yet highly effective. The **Project Manager (PM)** is **Marie Callanan**. She will be responsible for ensuring that the work programme adheres to its schedule, fulfils its objectives (according to the principles of on-time, on-budget and to high quality), and that the project meets the requirements of the company. The Project Milestones (Table 3.2 a) will be used as a valuable tool for determining whether or not the project is on schedule. She will be responsible for the management of the internal project development team, as well as in managing the relationship with stakeholders (beta clients) system from the VISUM existing client base) who will be involved at different stages of the work plan. She will be active in steering the evolution of the project by closely monitoring and controlling the progress of the work, as well as the engagements and contributions from the stakeholders. She will be responsible for ensuring that the project is managed in adherence with the internal project management procedures that are certified by ISO 9001:2008 and UNE-EN-ISO 166002: Management Systems for R&D&I, and with support from the VISUM Project Management Office (PMO). Marie will also be the interface with the sponsors (the European Commission) on behalf of VISUM, and she will ensure that all contractual obligations are fulfilled. Marie CV in Section 4.1 highlights her proven experience and track record in managing and participating in product developments, and her outstanding academic achievements whereby she graduated with a. B.Sc. in Engineering and went on MBA.

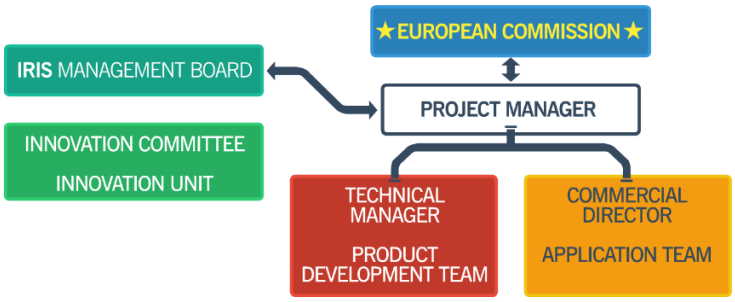

FIGURE 5- MANAGEMENT/ORGANISATIONAL STRUCTURE

The **VISUM Management Board, Chaired by General Manager, Mr. Colm Digby,** will be responsible for the strategic management of the project in line with the business and commercial strategy of the company. The VISUM Management Board will take all strategic decisions in relation to the project (e.g. decisions with regard to any major technical decisions, budgetary decisions, knowledge management/IP protection. Decisions will be taken following the normal decision-making procedures and voting rights of the VISUM Management Board. The PM will formally report the progress of the project to the Management Board every month and every quarter the budget spend will be reported to the board. Extraordinary meetings with the Management Board will be called as required to address, discuss and arrive at a swift resolution should any extraordinary problems or issues arise. Dr. Tim Kehoe, who is the Product Manager for the VISUM development, will lead the Product Development Team that will implement the work according to the project work plan. He will meet with the Project Manager and the development team on a weekly basis for steering technical discussions, monitoring the evolution of the work, measuring progress against the work plan, analysing potential risks and bottlenecks, contingency planning, etc. Issues that could have a strategic or critical impact on the project will be reported immediately to the General Manager, who will convoke an extraordinary meeting of the Management Board as appropriate. Oonagh Mc Nerney, the CEO, will lead the **Commercialisation team**, and she will be responsible for building the sales agent/distribution network internationally as well as further developing the business plan (with the inputs and approval of the Management Board). She will also coordinate with the marketing team to ensure the project dissemination and communication campaigns are aligned with the commercial strategy and she will lead the project's engagement with our group of 'beta clients'/early adopters. Together the Product Manager and the CEO will establish the Application Team that will provide technical support to the sales team.

■APPROPRIATENESS TO THE COMPLEXITY AND SCALE OF THE PROJECT: The VISUM® project will be managed following the same structure and decision-making mechanisms as an internal product development project within VISUM, and this is considered as a highly appropriate approach.

■INNOVATION MANAGEMENT: Innovation Management is a function that exists within the VISUM organisation structure and is part of our 'UNE-EN-ISO 166002: Management Systems for R&D&I' certification. The structure consists of an **Innovation Committee** (comprised of the General Manager, Director, Deputy Director and the Commercial Director), and an **Innovation Unit** (which is led by the Innovation Unit Head- a role that is occupied by the Science and Technology Manager in VISUM, along with senior scientists, researchers and engineers from the company). The Innovation Committee (which is an advisory body to the Management Board) meets with the Innovation Unit on a monthly basis in order to track the innovation potential of all active projects, to analyse IP, to discuss technology watch, state-of-the-art, competition, and market considerations around the R&D lines that the company is actively investing in. The Innovation Unit also assist the Innovation Committee in the development, maintenance and updating of the R&D Roadmap and the technology portfolio. The innovation management of VISUM will be performed by these Innovation Bodies within VISUM and will be managed through Task 6.3.

■CRITICAL RISKS: The critical risks (technology-, commercial-, and organisation-related) that could jeopardise this Innovation Project and appropriate risk-mitigation measures are provided in Table 3.2b. Through Task 6.2, a continual process of risk assessment and monitoring will ensure that the status of all risks is effectively managed and that action plans for all mitigation measures and contingency plans are implemented, monitored and reported. The process of risk identification and management will be based upon the principles and guidelines of the FERMA (ISO 31000 related) Standard[27].

3.3 CONSORTIUM AS A WHOLE (IF APPLICABLE)

Not applicable as this is a single entity project.

3.4 RESOURCES TO BE COMMITTED

TABLE 3.4A: SUMMARY OF STAFF EFFORT

Participant No.	Participant Short Name	WP1	WP2	WP3	WP4	WP5	WP6	Total Person/Months per participant
1	IRIS	18.0	96.0	29.0	29.0	22.0	14.0	208.0
Total		**18.0**	**96.0**	**29.0**	**29.0**	**22.0**	**14.0**	**208.0**

TABLE 3.4 B: 'OTHER DIRECT COST' ITEMS (TRAVEL, EQUIPMENT, INFRASTRUCTURE, GOODS AND SERVICES)

[27] Federation of European Risk Management Associations (www.ferma.eu)

1 IRIS	Cost (€)	Justification
Travel	33000	Travel to test sites to define integration strategies and requirements in WP1 (3K); Travel and subsistence for engineers for industry integration and demo work in WP3 (6k); Travel to meet suppliers, agents, distributors (12k); Travel to trade exhibitions and conferences (12k).
Equipment	20000	Software licenses
Other goods and services	482000	Materials (sensors, optomechanics, casing, etc.) for building 5 VISUM prototypes in WP2 (100K); Laboratory consumables (32k); IPR fees and patent costs (35K); Exhibition space and subscription to 4 tradeshows (50K); audit certificate (3K); Certification services (17K); Communication materials- video, leaflets, banners, posters, brochures (18K); Shipping and insurance (10K); Materials for constructing the manufacturing and assembly unit (90k); Construction of a demo rig (12k), Organisation of demo workshops (6k); Production of 5 VISUM product units (50K); Market reports (26K); Subscription to 4 conferences (8k); Cloud hosting fees (25k)
Total	535000	

4. MEMBERS OF THE CONSORTIUM

4.1 PARTICIPANTS

Participant no.	1	Name (short name)/Country	VISUM Ltd. (VISUM)/Ireland

Description of the legal entity

Visum Ltd.- www.visum.ie (previously known as IRIS Advanced Engineering Limited) is a Dublin-based technology company located in the campus of University College Dublin. The company was initially spun out as a branch facility of IRIS, S.L. from Spain (established in 2007 and with 53 staff) as part of a strategy to internationalize the company and establish an optical engineering division in Ireland. In 2013 this branch was converted into an indigenous Irish company.

VISUM is as an advanced engineering and R&D company, specializing in the field of process engineering. The facility in Dublin has its own team of optical, electronics and software engineers and has a fully equipped optics laboratory, mechanical and electronics workshop, and has solid expertise in developing spectroscopic and hyperspectral imaging solutions for industry (food, pharma and chemical advanced monitoring applications), as well as developing rigs for universities and research labs. We also have expertise in the development of cloud-enabled data management solutions for smart manufacturing and smart agriculture applications.

VISUM is **ISO 9001:2008 certified** and also is certified under **UNE-EN-ISO 166002: Management Systems R&D&i**. Since 2011, VISUM has a Commercial Department with its own sales and marketing team.

VISUM is a member of the following networks: SPIE (International Society for Optics and Photonics), SECPHO (Southern European Cluster of Photonics), SPIRE (Sustainable Process Industries).

Main tasks

VISUM Ltd. will fully implement the Phase 2 of the innovation project as the sole applicants. We have the technical and operational capacity and staff, as well as the research, engineering and product development capability, market knowledge and B2B commercial expertise that can be dedicated to this development. We will also expand our team in Dublin through new dedicated hires.

How the legal entity profile matches the tasks in the proposal

VISUM offers advanced monitoring and control turnkey solutions based on:

- Chemical monitoring (NIR)- handheld device and in-line solutions
- Hyperspectral imaging (foreign body detection and chemical composition analysis)
- Industrial cloud solutions and data management solutions for data processing, decision support, automation and data analysis (big production data)

We are engaged in R&D activities in the fields of optical and spectroscopy technologies (e.g. MEMs spectrophotometers, fluorescence and Raman) for process monitoring for on-line and PAT applications in food, pharmaceuticals and the process industries, as well as for controlling intensified processes.

IRIS has successfully developed pre-competitive IR spectroscopy prototypes in several FP7 projects (whereby, we have been responsible for integrating the sensors within the equipment and developing the optomechanics, in addition to the chemometric models for data processing, and the ad hoc software for supervision and control): VITISPEC –FTIR-ATR for monitoring wine composition-, OPTIMALT –NIR for real-time determination of barley protein and moisture in a harvester-, OPTIJECT –FT-NIR for monitoring plastic injection-, SPECTRAFISH –hyperspectral imager for fish QC-, OPTICLEAN –hyperspectral imager for CIP validation-, THIME –ellipsometer and hyperspectral imager for monitoring thin-film optoelectronic devices production lines.

We have successfully completed the Phase 1 SME Instrument VISUM (Grant Agreement- 684767), where we:

- Applied for a registered trademark for the VISUM brand
- Defined our brand positioning
- Secured the commitment of 4 first beta testing clients from the dairy, meat, olive oil and pharmaceutical sectors
- Defined the system specifications based on a Kano survey of the beta testers and the implementation of the Quality Function Deployment process
- Performed a technical feasibility study on fouling remediation technologies and defined the approach for going forward based on the results (i.e. a polymer coating option for in-line solutions and a sampling design for on-line solutions)

- Further developed our business plan and defined our IPR protection route
- Planned and costed the development work to get to product
- Participated in an Innovation diagnostic at Enterprise Ireland.

Thanks to the Phase 1 we have strengthened our knowledge of the market and the business opportunity, and we have a defined our product specs with a clear market positioning. As a company we are fully committed and invested in completing the VISUM platform and our intention is to be market ready for a product launch at ACHEMA 2018 in Germany.

CV of the persons who will be primarily responsible for carrying out the work

MARIE CALLANAN (FEMALE) Project Manager at Visum has an Electronic Engineering Degree from Trinity College, Dublin, and a Masters in Business Administration from the European University Barcelona. Marie comes from an engineering background with more than 15 years' experience in monitoring and control. She specialised in modelling system dynamics of engineering processes with particular expertise in wind, fuel cell, space and environmental applications. Marie has led and participated in many R&D projects in the fields of energy, environment, materials development and process monitoring. She has managed projects funded under national and international programmers and has expertise in best practices in the European Framework Programme project management going back to FP5. As a project manager Marie is adept at interpreting and communicating requirements, team building and risk management.

DR. TIMOTHY KEHOE (MALE) PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterisation of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials. Tim is the VISUM Product Manager and he brings vast optical integration and device instrumentation expertise.

DR. SEVERINE PHILIPPE (FEMALE) graduated with a 1st class Hons. B.Sc. in Applied Sciences (Physics and Physics Technology) from Dublin Institute of Technology, Ireland, in 2002 and received her PhD in Physics from Trinity College Dublin in 2007, for the study of polarisation effects in semiconductor optical amplifiers (SOA). She went on to work on the development and characterization of low noise SOAs, as well as THz generation by multimode lasers and their detection using four wave mixing in SOA, both in the Semiconductor Photonics Group, Trinity College Dublin, and the School of Electronic Engineering, Dublin City University. She previously worked as a R&D technician for the plastics industry and in quality control for electronic appliances. She joined VISUM in 2014 as an optical engineer and is currently involved in projects including various microscopy techniques, planar waveguide technologies and MIR spectroscopy. Serevine will be involved in the optical design work for the project.

NIKOLINA MIKULCIC (FEMALE) User Interface Designer. Nikolina comes from web design and development background with more than 15 years of experience in this field and she is an expert on UI/UX design strategy and front-end development (CSS/HTML). She studied graphic design and, because of her passion for the digital world, she studied web design. Since joining VISUM in 2016, she has been creating new User Interfaces for the VISUM product lines. Prior to VISUM, she has extensive experience in the online world, having worked in companies such as EuroArt93 – digital agency, BlueBOX IT, Biroval. She also was a founder of design studio In Love With Blue and she worked as Project Manager, UI/UX designer and front-end developer. She was leader on numerous digital projects for different clients such as Erste Bank, Unicredit Bank, Telenor Bank, Komb Bank and ZAMP.

MÉABH GARRICK (FEMALE) is an Optical Engineer with an M.Sc. Photonic from Friedrich Schiller Universität, Jena, Germany. Méabh has a B.Sc. in Physics with Medical Physics for the, National University of Ireland, Galway. She was awarded a scholarship to study at the Abbe School of Photonics, Friedrich Schiller Universität, Jena where she spent almost four years. She worked as a research scientist at the Fraunhofer IOF in Jena, Germany after completing her Master Thesis there. She specialised in the area of spectrally resolved laser-induced damage testing using a high-power tuneable OPO laser system as well as angle resolved scattering (ARS) from surfaces and the unique combination of both techniques. Her work was presented at the OSA Interference Coatings conference (June 2016). Méabh is the Technical Lead for two of VISUM's projects funded under H2020.

NEIL SHERIDAN (MALE) graduated from Dublin Institute of Technology, Ireland. in 2013 with a 2.1. B.Eng. (Mech). Neil has over 3 years engineering experience in various roles including Military Engineer, R&D Engineer, Automotive Production/Machine Design Engineer and Mechanical Engineer. He has strong abilities in CAD and creating innovative solutions which he has demonstrate in a number of projects. Neil was the sole mechanical engineer working on the mechanical design, testing, installation and validation for a major research project. He has also capabilities in electronics as shown in the optimisation of the PureFormula project which required significant rework to return it to working order. Neil is one of the main contributors on the mechanical design and build of the H2020 funded Visum Inline project. He has provided mechanical solutions in all aspects of the development from opto-mechanical systems to thermal control and integration into end users.

MARCELO SILVA (MALE) is a Front-end developer for Visum in Dublin. Completed his third level education in 2013 he was awarded a BA in Computer Science and Business from the National University of Ireland. Maynooth. In 2015 he was awarded an M.SC. in Cloud Computing by the National College of Ireland. Since 2014 he has worked as a software developer for Simteractive and Avaya.

CHRIS DERMODY (MALE) Software Developer. Bachelor's degree in Product Design from Dublin Institute of Technology. Chris excels in creative problem-solving, using the latest technologies to transform ideas into real opportunities that provide exponential value to business operations. With a keen eye for user experience, Chris builds robust, responsive applications that stand the test of time. Since joining VISUM, Chris has been hard at work developing both front and backend features that push the VISUM product lines forward. Prior to VISUM, Chris enjoyed working across the full spectrum of business, founding start-ups and overseeing product development throughout the lifecycle of many successful businesses.

EKAITZ HERNANDEZ TROYAS (MALE) Fullstack Developer at Visum. Bachelor Degree in Computing Science in the Public University of Navarra (5 years). Ekaitz comes from start-up background focusing in web developing. He has 4 years of experience in the web industry. He started to work in Dublin for a small start-up related with geolocation industry. He had the opportunity to create his own start-up related with the cosmetic industry. This experience has given him the opportunity to learn cutting edge technologies as well to learn other facets of a company like business development or marketing. Also, he worked as a freelancer implementing solutions with the Squarespace CMS. Ekaitz is always keen to face new challenges and learn new things out his core competency. He is currently developing several software development projects, including data management.

COLM DIGBY (MALE) General Manager and Co-Founder of VISUM. B.Sc. Hons. in Applied Computing and a Diploma in Industrial Studies from the University of Ulster, Magee College, N. Ireland. Colm is passionate about the Industry 4.0 paradigm with a connected and data driven everything. He is a strong leader and is driving the product development portfolio in IRIS. Prior to IRIS, Colm held positions at IBM Europe in Paris, the European Space Agency in Rome and Picis SA (and healthcare software provider). He has been instrumental in the digital transformation of the company in order to improve internal operations and efficiencies.

OONAGH MC NERNEY (FEMALE). Director and Co-founder of IRIS. She set up IRIS in 2007, with co-founder Colm Digby, with a vision to accelerate the transitioning of new technologies out of the lab and into industry. She is passionate about converting research and development into business. Over the years she has been a fundamental driver in the evolution of IRIS towards the development of spectroscopy based advanced monitoring solutions and in harnessing the power of process data to provide industry with knowledge and intelligence. With a background in Business, she is a marketing enthusiast and a believer in the power of digital marketing and branding to help companies to reach new global levels. Prior to IRIS, Oonagh spent 15 years working with industry, technology institutes and Business Development Agencies in the areas of technology transfer, innovation, business development and industry foreign direct investment. Expertise in new product development, leadership of applied R&D, commercialisation, branding and marketing.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Technical article "Injection moulding process optimization by online material monitoring". E. Bugnicourt et al., October 2012.
2. Guía para la evaluación sensorial de calidad de vinos tintos de Rioja Alavesa: vinos jóvenes y vinos con crianza en barrica (A guide to quality sensory evaluation of young red wines from Rioja Alavesa and barrel aged wines). I. Etaio, F.J. Perez-Elortondo, M. Albisu, J. Salmerón, M. Ojeda and E.Gaston. Central de Publicaciones del Gobierno Vasco. ISBN 978-84-457-2563-4
3. Technical paper, "Optimización del proceso de inyección mediante la monitorizacion en línea del material", E. Bugnicourt, V. Ruiz, O. Ghita, R. Ventura. Revista de plásticos modernos, Vol. 102. Número 660, 117-120, Septiembre 2011. Development and commercialisation of the VISUM instruments- NIR spectroscopy devices- in-pipe, handheld and Hyperspectral imaging- for real-time monitoring and quality control of raw materials, production lines, and final products.
4. Publication in relation to Innovation Management: **O. Mc Nerney**, A. Stachowicz, P. Czuprynski, E. Philpott, D. Kolman, D. Martin, E. Herrmann (2010). Strategic Guide to Successful Use and Dissemination of the Results of Research and Development Projects. 1st edition, Steinbeis-Edition Stuttgart 2010. ISBN 978-3-941417-27-4
5. On 08/02/2016, VISUM was awarded the Seal of Excellence form the European Commission (See Annex 3)

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

IRIS Advanced Engineering is participating in the following projects in H2020:

1. **PREVIEW**- PREdictiVe system to recommend Injection mold sEtup in Wireless sensor networks. (Grant Agreement number: 636892). IRIS is responsible for the development of the Data Acquisition System and in in developing the Signal Conditioning, its integration with the ARM main board, as well as implementing a prototype.
2. **VISUM**-An IR-spectroscopy-based platform for the rapid integration and customisation of affordable, real-time monitoring solutions for the food and pharmaceutical industry (Grant Agreement number: 684767). IRIS is the sole applicant of this SME Instrument project for accelerating the VISUM in-line NIR platform development and taking it to market.

3. Delivery of courses to industry in Process Analytical Technologies (PAT), e.g.: "Chemometric tools for process analytical technologies. ChemoPAT" University of Tarragona 16-18 July 2014; PAT: Innovando en tecnología de control de procesos farmacéuticos (Asociación de Químicos e Ingenieros del IQS)- May 28th 2015. **4.** In a B2B capacity IRIS Advanced Engineering Ltd has implemented the following private development contracts, among others, based on photonics e.g.: Development of an online polyphenols analyzer for real-time process control in presses for the wine industry; Development of a handheld NIR device for the at-line measurement of quality of oil in industrial frying equipment for bakery products (total acidity, polar compounds and degradation); Development of an ad hoc online end-point sensor for a novel blender for the pharmaceutical industry
Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
- Spectroscopic Laboratory equipped with several instruments: FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), 1 Extended-SWIR (StellarNet) and 1 NIR (Mightex) spectrophotometers as well as 1 SWIR Hyperspectral imager (Specim-Xenics). The associated staff are trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes). - Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the design of optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware"). - Wet chemistry and analytics lab - INNOVATION HUB with an accelerator room (for sprint work) demo and presentation area. Auditorium for more formal presentations and training.

4.2 THIRD PARTIES INVOLVED IN THE PROJECT (INCLUDING USE OF THIRD PARTY RESOURCES)

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	NO
Does the participant envisage that part of its work is performed by linked third parties	NO
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	NO

4.3 SUBCONTRACTING- BEST VALUE FOR MONEY

The best value for money principle of the subcontracts has been assessed during the evaluation. The Responsible Authorising Officer confirms that all the subcontracts of the proposal comply with the best value for money principle. For these subcontracts, beneficiaries have assurance that compliance with best value-for-money will not be challenged in audits — unless it turns out that the beneficiary did not follow the procedure described or concealed information for the purpose of the approval.

5. ETHICS AND SECURITY

5.1 ETHICS

During the preparation of this proposal, in keeping with the required Ethics Appraisal Procedure, an Ethics Self-assessment was performed with the completion of the Ethical Issues Table. It can be concluded that this project does not involve any ethics issues.

5.2 SECURITY

Does your project involve activities or results raising security issues	NO
Does your project involve 'EU-classified information' as background or results	NO

ANNEX 1- LETTERS OF INTENT

Almirall
Soluciones pensando en ti

Sant Andreu de la Barca, 22nd of July 2015

Industrias Farmacéuticas Almirall, S.A.
Crta. Nacional II, Km. 593
08740 Sant Andreu de la Barca
Barcelona-España
Tel. 93 291 30 00
Fax 93 291 33 44

Dear Ms. Granda,

With this Letter of Intent, I would like to express our interest to participate in the development of the VISUM platform in the framework of Horizon 2020 SME Instrument in order to communicate our industry requirements to assist in defining the specifications of the platform during the implementation of the Phase 1 of this project as well as to test the platform once it is built during the Phase 2 of the project.

We, INDUSTRIAS FARMACÉUTICAS ALMIRALL, S.A. (ALMIRALL), are the leading pharmaceutical company in R&D investment in Spain.

Our two manufacturing plants are located in Spain and Germany. These state-of-the-art production facilities are equipped with the most advanced technology.

Our history has seen many successes over the years, marked by consistent growth and a commitment to innovation: As we say at ALMIRALL, the best is yet to come!

We are convinced that process analytical technologies (PAT) such as proposed with the VISUM platform are an excellent tool to improve production efficiency.

Currently we use NIR devices for incoming material inspection but we have not integrated any NIR device in the production process in-line.

We are very happy to offer our facilities as test and implementation plant for an in-line VISUM platform in order to be one of the pioneers in the Spanish Pharmaceutical industry to use a PAT system in-line.

Yours sincerely,

Núria Busquets Baqué
Industrial Development Manager

Aceites del Sur - Coosur, S.A.
C.I.F.: ES A-80245129

Acesur
INNOVAMOS POR TRADICIÓN

Letterhead of company

Vilches,6th of October 2015

Dear Ms. Granda,

With this Letter of Intent, I would like to express our interest to participate in the development of the VISUM platform in the framework of Horizon 2020 SME Instrument in order to communicate our industry requirements to assist in defining the specifications of the platform during the implementation of the Phase 1 of this project as well as to test the platform once it is built during the Phase 2 of the project.

We, ACESUR, are one of the worldwide TP 5 leading companies in producing, packaging and marketing olive oil.

Our head office is in Dos Hermanas, Seville (Spain) and we have other facilities in Madrid, Toledo, Cuenca and Jaen (Spain). ACESUR covers all processes and markets within the oil sector.

Quality is the main goal of ACESUR. Our production process complies with the highest quality standard regulations, all our production processes are controlled by more than 20 analysts and we count on the most advanced managing techniques.

We are interested in IN-LINE MONITORING OF QUALITY PARAMETERS in olive oil, sauces and mayonnaises, to guarantee the quality of our products and to increase our production efficiency.

We are pleased to provide samples to conduct feasibility tests and to give access to our facilities in Vilches (Spain) to test the implementation of the NIR VISUM Inline system.

Yours sincerely,

Andrés García Sánchez
R&D Director

Calidad Pascual

Madrid, 16th of October 2015

Carretera de Palencia, s/n.
09400 Aranda de Duero.
Burgos. España.
T. +34 947 52 91 00
F. +34 947 52 91 11

Dear Ms. Granda,

With this Letter of Intent, I would like to express our interest to participate in the development of the VISUM platform in the framework of Horizon 2020 SME Instrument in order to communicate our industry requirements to assist in defining the specifications of the platform during the implementation of the Phase 1 of this project as well as to test the platform once it is built during the Phase 2 of the project.

We, CALIDAD PASCUAL, are one of the leading Spanish Dairy Companies.

We dream with a better future for everybody and every day we create new solutions. Our products are elaborated in plants with the most advanced technologies.

We believe in NIR technology because we already have long year experience to use NIR analyzers in laboratories.

Therefore we are happy to help in defining the requirements for a VISUM inline device and to give access to our plants to install a VISUM inline device in a first production line.

We are very interested in in-line measurement and in-line control of quality parameters of milk, yogurt and juices using NIR technology as offered with the VISUM In-line device. We consider this technology as an opportunity to increase at the same time our production efficiency and quality.

Yours sincerely,

Ignacio García-Cano Salgado
C.E.O., Director General Ejecutivo Calidad Pascual

Salgot
BUENOS EMPUTIDOS
Aiguafreda 1928

Embotits Salgot S.A.
Plaça Major, 14
08591 – Aiguafreda
Barcelona – Spain

Telephone +34 93 844 20 20
Facsimile +34 93 844 01 01
Email jpararedа@salgot.com
www.salgot.com

Aiguafreda, 21st of October 2015

Dear Ms. Granda,

With this Letter of Intent, I would like to express our interest to participate in the development of the VISUM platform in the framework of Horizon 2020 SME Instrument in order to communicate our industry requirements to assist in defining the specifications of the platform during the implementation of the Phase 1 of this project as well as to test the platform once it is built during the Phase 2 of the project.
We, EMBOTITS SALGOT S.A., a family business since 1928 elaborating sausages "Premium" believe in **solid traditions combined with an innovative spirit**. Thanks to this positioning our company has reached and international reputation of Premium Food brands in Spain and internationally.
The application of novel technologies such as NIR based in-line monitoring can **push SALGOT forward towards a new generation of quality control able to guarantee best quality together with highest production efficiency**.
We are very happy to test the new VISUM in-line device at our facilities and evaluate the results from the point of view of a MEDIUM SIZE COMPANY.

Yours sincerely,

Pere Valeri Salgot Nadal
Director
EMBOTITS SALGOT, S.A.
Conseller Delegat

ANNEX 2- BUSINESS MODEL CANVAS

Business Model Canvas. Designed for: VISUM® Inline . **Designed by:** Senior Management. **Date:** 06.04.16 **Version:** V1.4

Key Partners

Sensor providers:
- Spectral Engines
- Radiantis
- Infaimon

Key Activities

- Knowledge Generation to widen portfolio
- Product Development
- Chemometric models
- Product in-line integration
- Regulations and standards analysis
- Establish Application Team
- Training and materials for Sales Team
- Identify and arrive at agreements with distributors

Key Resources

- Optical sensors
- Optical, electronic and mechanical engineers
- Chemometrics knowledge
- Industrial integrators
- Technical commercial team
- The 'NIR Prof.'- Emil W. Cuirczak (acting as a technical, market and regulatory advisor)

Value Propositions

Food sector:

❑ **In-line quality analysis:**
- Chemical composition analysis
- Parameters quantification
- Particle size discrimination
- Adulteration identification
- Detection of contaminants

Pharma sector:

❑ **In-line quality analysis**
- Particle size discrimination

❑ **Process control and monitoring**
- Identification
- Blending

Customer Relationships

❑ **Commercial/Sales Team**
- Customer relationship development
- Satisfaction tracking
- Technical communication with client (assisted by Application team)

❑ **Application Team**
- Technical content for client communication
- Integration and technical support
- Chemometrics and customisation work

Channels

- Internal Commercial/Sales team
- Mailings (application notes)
- Social media (blogs, articles, etc.)
- Tradeshows
- Website (technical data: datasheets, application notes, etc.)
- Distribution network (to be established).

Customer Segments

❑ Food processing
- Beta clients (ACESUR, SALGOT, CALIDAD PASCUAL)
- Lighthouse clients (to be approached once beta tests are application notes): NESTLE, EUROPASTRY, TORRES, DAIRYGOLD

❑ Pharma processing
- Beta clients: ALMIRALL
- Lighthouse clients: SANOFI, GSK

❑ Other process industries (chemical)- DSM, P&G

Cost Structure

- Staff and overhead costs: Commercial/sales Team and Application Team (Life scientists, engineering and software/industrial cloud teams)
- Production and installation costs
- Distribution margin and sales agent margin
- Marketing

Revenue Streams

- Sale of VISUM in-line hardware
- Installation contracts
- Maintenance contracts
- Chemometric modelling and model maintenance
- Data services- potentially as a Subscription model (SaaS)- monthly subscription and 'pay per use'

http://www.businessmodelgeneration.com

ANNEX 3- SEAL OF EXCELLENCE

SEAL OF EXCELLENCE

European Commission

Certificate delivered by the European Commission,
as the institution managing Horizon 2020,
the EU Framework Programme for Research and Innovation 2014-2020

The project proposal **719044, VISUM**

An IR-spectroscopy-based platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for the food and pharmaceutical industry.

Submitted under the Horizon 2020's **SME instrument phase 2**
call **H2020-SMEInst-2014-2015 (H2020-SMEINST-2-2015)** of **25 November 2015**
in the area of
Accelerating the uptake of nanotechnologies, advanced materials or advanced manufacturing and processing technologies by SMEs
by
IRIS ADVANCED ENGINEERING LIMITED
COOLKERAGH LOWER
LISTOWEL KERRY
Ireland

following evaluation by an international panel of independent experts

WAS SUCCESSFUL IN A HIGHLY COMPETITIVE EVALUATION PROCESS*
AS AN INNOVATIVE PROJECT PROPOSAL

This proposal is recommended for funding by other sources since Horizon 2020 resources available for this specific Call were already allocated following a competitive ranking.

* This means passing all stringent Horizon 2020 assessment thresholds for the 3 award criteria (excellence, impact, quality and efficiency of implementation) required to receive funding from the EU budget Horizon 2020.

Corina Crețu,
Commissioner for
Regional Policy

Carlos Moedas,
Commissioner for Research,
Science and Innovation

Brussels, 08/02/2016